The truncated Weil quadratic form: inertia, visibility, structural measurements, and no-go theorems for the Riemann Hypothesis

Logan Eberwein · Independent Researcher · loganeberwein@icloud.com · ORCID 0009-0004-5650-9313 · October 2026 · Supersedes and extends the author's progress report (doi:10.5281/zenodo.23174797) · Code, data and supplementary ledger: github.com/loganeberwein/Logan-Eberwein-RH-Weil-Window · Lean 4 sources (§12): github.com/loganeberwein/Logan-Eberwein-RH-Research (branch stage31-two-channel)

Abstract

We record observations on the truncated (window) Weil quadratic form that, to our knowledge, are not stated explicitly in the literature as of October 2026, together with negative results and retracted claims. None of this proves or approaches a proof of the Riemann Hypothesis. The aim is to save future researchers (human or AI) from re-deriving these connections, and to give a clear picture of what does not work: which routes fail, and exactly why.

The new content:

1. An inertia identity counting the negative directions of the window form through a 2×2 "fiber matrix" attached to the two poles of ζ (§2; finite-dimensional theorem).
2. **No-go theorems with proofs (§7):** a blind zone uniform in height (any off-line quadruple shallower than $\delta_0(L) \approx \sqrt{(3\mu(L)/L^3)}$ is invisible to a window of length L, at every height), an exact height-dependent invisibility criterion, a Krein-type indistinguishability corollary, a finite-jet no-go, the Cauchy–Schwarz inflation obstruction to an unconditional Conrey–Ghosh–Gonek argument, the Beurling-blindness of Selberg-identity arguments, a two-sided fake-world barrier, the non-existence of a Selberg-class tensor square, and the absence of log p from arithmetic geodesic length spectra.
3. The Connes–Consani–Moscovici ground state is reproduced to angle $1 - 10^{-15}$ by one linear solve against the pole function (§3).
4. The window's reproducing kernel equals 1.000000 at every in-band zeta zero and 10^8–10^{37} between zeros (§4).

5. A visibility function $\delta^*(L, T)$ with an empirical band edge near $4 \cdot X \cdot \log X$, and a per-zero accuracy law $e_k \approx c_k \cdot \delta^*(\gamma_k)^2$ with window-independent constants (§5–6).
6. **Structural measurements (§8):** positivity is all-or-nothing over the primes (no proper team of primes is self-sufficient, by order-one margins); the archimedean part is indefinite and the primes repair it; for curves over finite fields the near-null space of ζ is an *exact* radical, so ζ''s knife-edge is a blurred exact radical; and the renormalized margin of a doomed world rises before it fails, which rules out scalar flow arguments.
7. A catalogue of dead routes with precise kill reasons: nine direct proof attempts, thirteen absolute-arithmetic routes, a ten-direction parallel search for a gluing principle, and the price every standard detector pays in scale or precision (§10).
8. Negative results of the geometric dictionary (§9), further observations (§11), the SR/Rees matrix program with its Lean-verified theorems, exact failure theorems for specific routes, structural obstructions from about 150 research cycles, the six-property operator criteria, conjectures, and corrections to the author's earlier report (§12), and retracted claims (§13).

Status labels used throughout: **Theorem** (proved here or classical), **Numerical** (computed, not certified), **Heuristic**, **Negative**, **Retracted**.

Authorship and AI-assistance statement: this is a year-long research program begun by the author early in his freshman year of high school. The author set the research directions; computation, literature checking, Lean formalization and drafting were carried out with AI tools (Anthropic's Claude, including Claude Code; OpenAI's GPT and Codex models), which are listed as tools, not authors. Numerical claims are reproducible from the stated formulas, the reference values in §1 and the scripts in the companion repository (see its README for the section-by-section map).

1. Setup and normalization

We use test functions f on the window $u \in [0, L]$, $L = \log X$ ($X = \lambda^2$ in Connes–Consani–Moscovici notation). With autocorrelation $q(y) = \int f(u)\, f(u+y)\, du$, the window form is

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W(f)=\int_0^L\Big[2\cosh(y/2)\,q(y)-\frac{e^{y/2}q(y)-q(0)}{2\sinh y}\Big]dy-
c_0\,q(0)-\sum_{p^m\le X}\frac{\log p}{p^{m/2}}\,q(m\log p),
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with $c_0 = (\log 4\pi + \gamma)/2 - \tfrac{1}{2}\cdot\log((X+1)/(X-1))$. This is one half of the usual explicit-formula sum: each zero pair $\pm\gamma$ contributes $|\hat{f}(\gamma)|^2$, where $\hat{f}(t) = \int f(u)e^{itu}\, du$.

We split $W = P - I$, where P is the pole term (the $2\cosh(y/2)$ part, rank two, signature $(1,1)$) and $I = P - W$ is the archimedean-plus-prime part. Writing $P = a\,b^{\mathsf T} + b\,a^{\mathsf T}$, the vectors a, b are proportional to the pairings with $e^{-u/2}$ and $e^{u/2}$.

In the Fourier basis $U_n = e^{2\pi i n u/L}/\sqrt{L}$, every matrix entry reduces to $2N+1$ scalars, as in Connes–van Suijlekom (Prop. 4.1): with $A_k = \Psi(\sin(2\pi k y/L))$ and $B_n = \Psi(2(1 - y/L)\cos(2\pi n y/L))$, where Ψ is the functional above applied to q,

$$W_{nn}=B_{|n|},\qquad W_{nm}=\frac{A_m-A_n}{\pi(n-m)}\ (n\ne m),\qquad A_{-k}=-A_k.$$

Cosine and sine blocks decouple exactly (even and odd under $u \mapsto L - u$). In arb arithmetic at 400–700 bits the assembly takes seconds for $X \leqslant 15$.

Reference values ($X = 3$, independently reproduced by a second implementation to 40+ digits):

quantity	value
A_1	−0.4134682699530559220560045874253097409382
A_5	−1.187044256530619490296122772175793667866
B_0	0.122328629825014959399377519059923110848 3
B_1	0.153193452967582144323644129782391676496 8
B_5	1.311521329536738919487937865214365776025
lowest eigenvalue, cosine block, $\lvert n\rvert \leqslant 40$	$5.757397320790458275680582968 09 \times 10^{-8}$
lowest eigenvalue, sine block, $1 \leqslant n \leqslant 40$	$1.707720679523380639921420100 93 \times 10^{-5}$

Fourier-basis eigenvalues are Galerkin upper bounds; the converged value at $X = 3$ is about 5.5319540×10^{-8} (graded spectral elements).

2. Result 1: an inertia identity through a 2×2 fiber matrix (Theorem)

Let $\nu(\cdot)$ and $\pi(\cdot)$ denote the numbers of negative and positive eigenvalues. If I is invertible, set $\alpha = a^{\mathsf T} I^{-1} a$, $\beta = b^{\mathsf T} I^{-1} b$, $\gamma = a^{\mathsf T} I^{-1} b$ and

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S=\begin{pmatrix}\alpha & \gamma-1\\ \gamma-1 & \beta\end{pmatrix},\qquad
\nu(W)=\pi(I)+\nu(S)-1.
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Proof. With U = [a b] and J the 2×2 swap matrix, the block matrix K = [[−I, U], [U^T, −J]] has Schur complements W (eliminating −J) and −J + $U^T I^{-1} U$ = S (eliminating −I). Haynsworth inertia additivity applied both ways gives the identity. ∎

Corollaries.

- W ⩾ 0 forces π(I) ⩽ 1, since I = P − W and P has one positive eigenvalue (Weyl).
- Window positivity is equivalent to π(I) ⩽ 1 together with S ⪰ 0; by the functional-equation symmetry α = β, so the second condition is the scalar inequality Δ_L = α − |1 − γ| ⩾ 0.
- The archimedean-plus-prime part I plays the role of the intersection form in Weil's proof for curves, and P the role of the fiber-degree product $2(D\cdot F_1)(D\cdot F_2)$: W is the Castelnuovo–Severi expression, π(I) ⩽ 1 is the Hodge-index statement, and S encodes the fiber plane.

Numerics. The identity was checked at X ⩽ 3.6 for ζ and for perturbed models in which the p = 2 terms are rescaled (by 0.5, 1.3, 2). In every perturbed model positivity fails through π(I) > 1, never through S alone. For ζ the smallest eigenvalue of S tracks the lowest eigenvalue of W, with ratio moving smoothly from 1.51 (X = 1.5) to 0.74 (X = 3.6). The Hodge margin (second eigenvalue of I) is 10^4–10^5 times larger than the margin of W at X ≈ 3–3.6.

Relation to prior work. The Castelnuovo–Severi analogy for Weil's form goes back to Connes (1999) and Connes–Consani–Marcolli. A parity-sector version of the same structure appears in a June 2026 note (Zenodo 20694588): there the pole term is 2|C⟩⟨C| − 2|S⟩⟨S| with C = cosh(x/2), S = sinh(x/2), the even pole-free operator has a single negative eigenvalue, and ground states are roots of rank-one secular equations. Our identity is the non-sectored statement, with the Hodge count and the fiber matrix separated, and the observation that ζ's knife-edge sits in S while perturbed models fail in π(I).

Caution. By the explicit formula W = Z (the zero sum), so I = P − Z. An off-line zero pair adds an indefinite block to Z and hence a positive direction to I. The Hodge-index condition is therefore RH-strength content in different coordinates, not a shortcut.

3. Result 2: the ground state is one linear solve against the pole function (Numerical, exact explanation)

Let p be the coefficient vector of the centered pole function $\cosh((u - L/2)/2)$. In the cosine basis $C_0 = 1/\sqrt{L}$, $C_n = \sqrt{(2/L)}\cdot\cos(2\pi n u/L)$, with $\omega_n = 2\pi n/L$,

$$p_n=\mathrm{nrm}_n\cdot\frac{\sinh(L/4)}{\omega_n^2+1/4},\qquad h=W^{-1}p\ .$$

Why h is the ground state. Take $H = I^{-1}(a + b)$, the "ample class" of the Castelnuovo–Severi dictionary. Then $W\cdot H = (\alpha + \gamma - 1)(a + b)$, and $\alpha + \gamma - 1$ is the small eigenvalue of S. So H is proportional to $W^{-1}(a + b)$, and W^{-1} applied to any vector is dominated by the lowest eigenvector, with relative error of order μ_1/μ_2 (ratio of the two smallest even eigenvalues).

Numerics (600-bit arb, Fourier basis):

X	N	μ_1 (even block)	$1 - \cos\angle$(h, eigenvector)	zeros matched
6	75	8.48×10^{-23}	9.1×10^{-14}	14 of 14 below T = 62
10	155	1.66×10^{-43}	1.5×10^{-15}	43 of 44 below T = 132

The zeros of $\hat{h}$ match those of the eigenvector's transform to all printed digits, and are marginally more accurate at low zeros (X = 10, γ_1: 9.07×10^{-40} for h vs 9.69×10^{-40} for the eigenvector).

Positivity. h(u) is strictly positive inside the window and vanishes only at u = 0 and u = L (X = 3, 4, 6, 10; min/max = 1.5×10^{-21} at the ends for X = 10). The same holds for $H = F_1 + F_2$, the sum of the two fiber classes $F_1 = I^{-1}a$, $F_2 = I^{-1}b$, while each F_i separately changes sign (about 48% of the window negative).

Relation to prior work. Connes–Consani–Moscovici (arXiv 2511.22755) use the eigenvector of the smallest eigenvalue. The rank-one secular equation of Zenodo 20694588 gives the even ground state exactly as $(B_e - \lambda_0)^{-1}C$; h is its zero-shift approximation, which needs no eigenvalue. Perron positivity of the ground state is discussed in a companion note ("The pole term is the only obstruction to Perron structure"). Reality of the eigenvector's zeros is proved by Connes–van Suijlekom (arXiv 2511.23257, Thm. 6.1). What we add: the closed-form right-hand side, the Theorem-A explanation of why the approximation is exponentially good, and the measured agreement. Reality of the zeros of $\hat{h}$ itself is not proved.

4. Result 3: the window kernel equals 1 at every in-band zero (Numerical; classical mechanism)

On a positive window, $K(t, t) = c(t)^{H} W^{-1} c(t)$, where $c(t)$ is the vector of $\hat{f}(t)$ over the basis, is the reproducing kernel of the window space normed by W.

Bound. Under the explicit formula with zeros on the line, $W(f) = \Sigma_{\gamma>0} |\hat{f}(\gamma)|^2 \geqslant |\hat{f}(\gamma_k)|^2$ for every zero, so $K(\gamma_k, \gamma_k) \leqslant 1$.

Measured saturation ($X = 10$, $N = 150$, 600 bits):

zero	K at the zero	K midway to the next zero
$\gamma_1 = 14.135$	1.000000	1.0×10^{37}
$\gamma_2 = 21.022$	1.000000	5.0×10^{33}
$\gamma_6 = 37.586$	1.000000	2.8×10^{25}
$\gamma_{11} = 52.970$	1.000000	3.2×10^{18}
$\gamma_{21} = 79.337$	1.000000	1.8×10^{8}
$\gamma_{29} = 98.831$	1.000000	2.6×10^{2}
$\gamma_{34} = 111.030$	0.999096	0.60
$\gamma_{39} = 121.370$	0.975736	0.80

In words: inside the band the zeta zeros are exactly the points where pointwise evaluation is weakest, and the bound is attained to six digits; past the band edge the pattern dissolves.

Interpretation. This is the classical fact that Christoffel functions converge to the mass of an atom (Freud–Nevai theory), applied to the zero measure $\Sigma\, \delta_\gamma$ (unit masses in this normalization). It is a consequence of the exact zero-sum identity for the truncated form (Groskin, arXiv 2607.02828). An off-line pair would invalidate the bound $K \leqslant 1$ at its height, which suggests K as a pointwise diagnostic. We did not find this saturation stated for the window Weil form.

5. Result 4: the visibility function δ*(L, T) (Numerical)

Definition. Planting an off-line quadruple at height T and depth δ (zeros ½ ± δ ± iT) adds, to second order on test functions with f̂(T) = 0, the term $-2\delta^2|\hat{f}'(T)|^2$ to W (in the normalization of §1). The window flags the pair exactly when δ exceeds

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\delta^*(L,T)^2=\min_{\hat f(T)=0}\frac{W(f)}{2\,|\hat f'(T)|^2}.
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This is a 2×2 generalized eigenproblem after restricting to the codimension-2 subspace f̂(T) = 0. Computing it from the prime-side matrix (which equals the sum over all zeros) avoids truncating any zero list.

Exact check. Planting the exact quadruple form instead of its second-order expansion and bisecting for the first negative eigenvalue agrees with δ* to 0.05% (X = 4: 9.3665×10^{-4} both ways at T = 5; 3.8039 vs 3.8060 $\times 10^{-2}$ at T = 18).

Values (Fourier-basis upper bounds, converged to 1–3% between N = 100/150 and 130/180):

X = e^L	T = 5	T = 18	T = 50	T = 75	T = 100	T = 150
4	9.2×10^{-4}	3.7×10^{-2}	2.3	3.1	3.1	3.5
6	2.5×10^{-8}	2.1×10^{-6}	0.23	1.8	1.8	2.4
8	3.3×10^{-13}	3.7×10^{-11}	2.8×10^{-3}	0.43	1.4	1.6
10	3.2×10^{-18}	4.3×10^{-16}	1.3×10^{-7}	2.5×10^{-3}	1.07	1.2
12	2.6×10^{-23}	4.0×10^{-21}	2.8×10^{-12}	2.2×10^{-7}	4.5×10^{-3}	0.90

Entries above 0.5 mean the window is blind at that height (real zeros have δ ⩽ ½).

Empirical laws.

- In-band sensitivity: at fixed T, $\log_{10} \delta^*$ drops by 2.27–2.38 per unit of X (X = 4 to 15), i.e. δ^{*2} falls about 5 decades per unit X, matching the decay of the lowest eigenvalue (consistent with the prolate-leakage heuristic of Connes 2026, §6.4).
- Band edge $T_b(X)$, the height where δ* first reaches ½: 23, 50, 68, 95, 117, 167 at X = 4, 6, 8, 10, 12, 15. The ratio $T_b/(X \log X)$ is 4.2, 4.7, 4.1, 4.1, 3.9, 4.1. The X log X form is a fit on six points, not established.
- Above the band, $\delta^* \cdot L^2$ ranges over roughly 6–10; an earlier "$\delta L^2 \approx 6$" detection rule is only this out-of-band regime.

Relation to prior work. arXiv 2608.24827 shows certification must resolve frequencies up to a doubly exponential threshold and that the margin collapses at the Landau–Widom rate. The band edge here is plausibly the Landau–Widom plunge region combined with zero density; that derivation is open. **Caveat:** δ* is a counterfactual (an extra pair planted into the actual world); a certified window cannot be converted into an exclusion region for actual off-line zeros, because the pair would already be present in the computed form and the conversion cancels at order δ^2.

6. Result 5: per-zero accuracy law $e_k \approx c_k \cdot \delta^*(\gamma_k)^2$ (Numerical)

Let e_k be the distance from γ_k to the nearest zero of $\hat{h}$ (§3). Then $e_k/\delta^*(X, \gamma_k)^2$ is nearly constant over 35 orders of magnitude, and the constant belongs to the zero, not the window:

zero	c_k at $X = 6$	c_k at $X = 10$
$\gamma_1 = 14.135$	0.144	0.142
$\gamma_4 = 30.425$	0.073	0.068
$\gamma_7 = 40.919$	0.061	0.052
$\gamma_{10} = 49.774$	0.059	0.044
$\gamma_{16} = 67.080$	(past band)	0.037
$\gamma_{25} = 88.809$	(past band)	0.036

At $X = 10$ the raw values run from $e_1 = 9.07 \times 10^{-40}$ ($\delta^* = 7.98 \times 10^{-20}$) to $e_{31} = 1.14 \times 10^{-3}$ ($\delta^* = 0.154$); the law breaks down above $\gamma \approx 105$, just past the band edge, and the approximant stops tracking zeros near $\gamma \approx 110$–120 (about 1.2 T_b).

Heuristic. Moving an on-line zero is a first-order effect on W, while displacing a zero off the line is second-order (an off-line pair at depth δ acts like an on-line pair with imaginary gap 2iδ). A window that detects depth δ* therefore pins positions to about δ^{*2}. The constants c_k decrease with height roughly like the local zero spacing; this was not tested further.

Relation to prior work. Groskin (arXiv 2605.20224) found the first-zero error and the smallest eigenvalue moving in lockstep in log space across 113 orders of magnitude (ratio of logs within 0.4% of 1), and the first ten zeros converging at rates within 3.8% of each

other. The per-zero law above is a refinement in different coordinates: it predicts each zero's error from a quantity computable at that height, with a window-independent constant.

7. No-go theorems (with proofs)

Each statement below has the form "a method that sees only X cannot decide RH" or "this route needs an input equivalent to RH". Several turn walls hit during this program into formal statements. Notation as in §1: a *world* Z is a set of spectral points $\gamma = (\rho - \tfrac{1}{2})/i$, symmetric under $\gamma \mapsto -\gamma$ and $\gamma \mapsto \bar{\gamma}$; on-line zeros have real γ. For real f on a window of length L (centered at 0, which is harmless because the form is shift-invariant), $W_Z(f) = \tfrac{1}{2}\,\Sigma_{\gamma\in Z}\, \hat{f}(\gamma)\cdot\overline{\hat{f}(\bar{\gamma})}$. An off-line quadruple at height T and depth δ is $Q = \{\pm T \pm i\delta\}$.

7.1 A blind zone uniform in height (Theorem)

Theorem 7.1. If $W_{Z_0}(f) \geq \mu\|f\|^2$ on window L, then for every height T,

$$W_{Z_0\cup Q}(f)\ \ge\ \big(\mu-\kappa(\delta,L)\big)\,\|f\|^2,\qquad \kappa(\delta,L)=\frac{2\sinh(\delta L)}{\delta}-2L=\frac{\delta^2L^3}{3}+O(\delta^4L^5).$$

Proof. For real f put $a = \hat{f}(T - i\delta)$, $b = \hat{f}(T + i\delta)$. Since $\hat{f}(-\bar{z}) = \overline{\hat{f}(z)}$, the quadruple contributes exactly $2\cdot\mathrm{Re}(a\,\bar{b}) = |a|^2 + |b|^2 - |a - b|^2 \geq -|a - b|^2$. Now $a - b = 2\int f(u)\, e^{iTu} \sinh(\delta u)\, du$, so by Cauchy–Schwarz $|a - b|^2 \leq 4\|f\|^2 \int_{-L/2}^{L/2} \sinh^2(\delta u)\, du = \kappa(\delta, L)\|f\|^2$. ∎

Consequences. Let $\delta_0(L)$ solve $\kappa(\delta_0, L) = \mu(L)$, so $\delta_0 \approx \sqrt{(3\mu(L)/L^3)}$. Adding an off-line quadruple of depth below δ_0 *at any height* leaves the window form positive definite. Since $\mu(L)$ decays like the prolate leakage, roughly $e^{-4\pi X}$ with $X = e^L$, one gets $\delta_0 \approx e^{-2\pi X}$: a window cannot see depth δ at any height until $X \gtrsim (2\pi)^{-1} \ln(1/\delta)$ (heuristic in the exponent, rigorous given μ).

Unconditional instance. arXiv 2608.24827 certifies a coercive bound of order 8.9×10^{-18} for support 1.6. Applied with Theorem 7.1, every hypothetical off-line quadruple of depth below about 2×10^{-9}, at any height, leaves that window positive (the constant depends on matching their normalization to ours, a factor of 2).

7.2 A height-dependent criterion (Theorem)

Theorem 7.2. Let W_{Z_0} be positive definite on window L and $h_T(u) = e^{-iTu} \sinh(\delta u)$. Let λ_T be the largest value of $|\langle f, h_T\rangle|^2 / W_{Z_0}(f)$ over real f (a 2×2 eigenvalue problem built from $W_{Z_0}^{-1}$ applied to the real and imaginary parts of h_T). If $\lambda_T \leqslant ¼$ then $W_{Z_0 \cup Q} \geqslant 0$.

Proof. From the proof of 7.1 the quadruple contributes at least $-|a - b|^2 = -4|\langle f, h_T\rangle|^2 \geqslant -4\lambda_T W_{Z_0}(f)$. ∎

Numerics (Z_0 = ζ′s zeros via the prime-side matrix, 600-bit arb, Fourier basis; small-δ linearization $h_T \approx \delta \cup e^{-iTu}$):

X	T	guaranteed invisible below	exact second-order threshold δ* (§5)	ratio
6	5	4.3×10^{-11}	2.6×10^{-8}	0.002
6	50	1.9×10^{-2}	0.23	0.08
6	100	1.30	1.83	0.71
10	5	1.8×10^{-21}	3.2×10^{-18}	0.001
10	75	4.0×10^{-6}	2.5×10^{-3}	0.002
10	150	0.79	1.20	0.66

Above the band edge the guarantee exceeds ½, the largest possible depth, so every off-line quadruple at those heights is invisible to that window. Exact form: evaluating λ_T with the exact $h_T(u) = e^{-iTu} \sinh(\delta(u - L/2))$ (no linearization) reproduces the small-depth rows to four digits (4.2888×10^{-11}, 1.9249×10^{-2}, 3.9693×10^{-6}) and gives, at $\delta = ½$, λ_T = 0.0390 (X = 6, T = 100), 0.0246 (X = 6, T = 150), 0.110 (X = 10, T = 150) and 0.0392 (X = 10, T = 200), all below ¼; so at those heights every off-line quadruple of every possible depth is invisible to that window. These values change by less than 0.3% between basis sizes N = 100/150 and 150/200. Remaining caveat: a Galerkin subspace yields a lower bound for λ_T, so the margins (a factor 2 to 10 below ¼) are strong numerical evidence, not a certificate.

7.3 Window tests cannot distinguish (Corollary)

Corollary 7.3. If $W_{Z_0 \cup Q} \geqslant 0$ on window L, then by Krein′s extension theorem the windowed data are the restriction of the Fourier transform of a positive measure ν, a world with entirely real spectrum. $Z_0 \cup Q$ (off-line quadruple) and ν (none) then have

identical prime-side data up to e^L, so any procedure reading only Λ(n) for n ⩽ e^L plus the archimedean and pole terms returns the same verdict on both.

Scope. ν is a measure, not necessarily the zero set of an L-function, and the statement is about planted worlds. Krein proved the extension theorem for continuous positive definite functions on an interval (Dokl. Akad. Nauk SSSR 26 (1940)); the extension problem for positive-definite distributions on an interval is treated in Gabardo, Mem. Amer. Math. Soc. 489 (1993).

7.4 Certified windows cannot be converted into exclusion regions (Proposition)

If ζ had an off-line quadruple, it would already be present in the computed window form W = W_{rest} + (quadruple). On test functions with f̂(T) = 0 the quadruple shifts the quotient W(f)/(2|f̂′(T)|²) by exactly −δ² to second order, so any bound on δ derived from a certified W cancels itself. Thresholds such as δ*(L, T) are therefore counterfactual (an *additional* pair planted into the actual world), never exclusion statements about actual zeros.

7.5 Finite analytic snapshots cannot force real-rootedness (Proposition)

Proposition 7.5. Let a test inspect finitely many values and derivatives Ξ^{(k)}(z_j), j ⩽ m, k ⩽ r_j. For every non-real w there is an entire function $\widetilde{\Xi}$ of order 1 with $\widetilde{\Xi}(-z) = \widetilde{\Xi}(z)$, $\widetilde{\Xi}(\bar{z}) =$ conj $\widetilde{\Xi}(z)$, identical inspected jets, and zeros at w, −w, w̄, −w̄.

Proof. By Hermite interpolation choose a polynomial P with P = 1 to order r_j at every ±z_j, ±z̄_j and P(±w) = P(±w̄) = 0. The symmetrization R(z) = ¼[P(z) + P(−z) + conj P(z̄) + conj P(−z̄)] satisfies the same (symmetric) conditions and is even and real. Put $\widetilde{\Xi} = \Xi \cdot R$. ∎ (Folklore in spirit; recorded as the analytic counterpart of 7.1.)

7.6 Cauchy–Schwarz inflation: the obstruction to an unconditional Conrey–Ghosh–Gonek argument (Proposition)

Conrey–Ghosh–Gonek and Bui–Heath-Brown obtain, under RH, a proportion 19/27 of simple zeros from N* ⩾ |Σ c_ρ|²/Σ |c_ρ|², c_ρ = ζ′(ρ)M(ρ). Without RH the denominator becomes the holomorphic pairing Σ c_ρ·conj(c_{ρ*}), ρ* = 1 − ρ̄, which is computable by contour integration but is no longer a sum of squares.

Proposition 7.6. An off-line pair contributes s + t to the numerator sum and 2·Re(s t̄) to the pairing, where s = c_ρ, t = c_{ρ*}. With t = s e^{iφ} and φ → π/2⁻, the ratio |s + t|²/(2 Re(s t̄)) = |1 + e^{iφ}|²/(2 cos φ) → ∞. Hence no inequality of the form N_{on,simple} + C·N_{off} ⩾ |Σ c_ρ|²/Σ c_ρ conj(c_{ρ*}) holds with an absolute C. ∎

An unconditional version therefore needs an additional estimate controlling the phases of c_ρ/c_{ρ^*} (for instance a mollified discrete fourth moment of ζ'). In Montgomery's pair-correlation setting this is supplied by the fixed exponential feature vectors (each zero tied to its mirror with a normalization; compare Lamzouri, arXiv 2609.02882, Prop. 2.1), which is why that method could be made unconditional.

7.7 Selberg-identity arguments are blind to addition (Proposition)

Proposition 7.7. For every Beurling generalized number system with zeta function $\zeta_B = \Pi_p (1 - p^{-s})^{-1}$, the identity $(\zeta_B'/\zeta_B)' + (\zeta_B'/\zeta_B)^2 = \zeta_B''/\zeta_B$ holds, i.e. $\Lambda_B(n) \log n + (\Lambda_B * \Lambda_B)(n) = (\mu_B * \log^2)(n)$. Diamond–Montgomery–Vorhauer constructed Beurling systems satisfying the prime number theorem whose zeta functions have zeros arbitrarily close to $\mathrm{Re}\, s = 1$. Hence no argument whose inputs are formal Euler-product identities (Selberg's included), the prime number theorem and positivity of Λ can prove RH. ∎ In particular a "completion of squares" built from Selberg's identity cannot close the gap; the additive structure (functional equation) must enter at the same order.

7.8 The two-sided barrier (Proposition)

world	keeps	loses	RH analogue
Davenport–Heilbronn	functional equation, Dirichlet-series shape	Euler product	false
Beurling systems (Diamond–Montgomery–Vorhauer)	Euler product, prime number theorem	additive structure, functional equation	can fail
Epstein zeta, class number > 1	functional equation, sum of Euler products	single Euler product	false
Curves over finite fields	both	—	true (Weil)

A valid proof must use a single Euler product and the additive structure together; any argument whose hypotheses hold in one of the first three rows is invalid.

7.9 No tensor square in the Selberg class (Proposition)

Deligne's tensor-power trick would need, for $k = 2$, a function L_2 whose zeros include all sums $\rho_1 + \rho_2$ (suitably normalized) of zeros of ζ.

Proposition 7.9. (a) If conjugate pairs are included, $\rho + \bar{\rho} = 2 \operatorname{Re} \rho$ gives infinitely many zeros in a bounded real segment, so $L_2 \equiv 0$. (b) If only zeros in the upper half-plane are used, the number of sums with imaginary part $\leqslant T$ is of order $(T \log T)^2$, whereas every function in the Selberg class has $N(T) \asymp T \log T$. Hence no Selberg-class L_2 exists. ∎

The underlying reason: closed loops of lengths log p and log q close up together only when p^a = q^b, i.e. p = q, so unique factorization kills the off-diagonal part of any would-be square of Spec $\mathbb{Z}$.

7.10 Arithmetic geodesic lengths are never log p (Proposition)

Proposition 7.10. Let $\Gamma \subset \mathrm{SL}_2(\mathcal{O}_K)$ be an arithmetic Fuchsian group. A hyperbolic $\gamma \in \Gamma$ has eigenvalues λ, λ^{-1} that are algebraic integers with product 1, hence units, and the closed geodesic has length $2 \log|\lambda|$. Since a unit has norm ±1 while p^m has norm $p^{m[K:\mathbb{Q}]}$, $|\lambda|^{2k}$ is never a rational prime power. So the length spectrum of such a surface never reproduces the prime-power comb $\{m \log p\}$, and a Selberg trace formula on it cannot be Weil's explicit formula. ∎ Allowing S-units makes log p available, but the multiplicative action of the primes in S on $\mathbb{R}_{>0}$ has dense orbits ($\log p/\log q$ irrational), destroying the discreteness a trace formula needs.

7.11 The heat flow has no real-rooted ancestor (Proposition)

Rodgers–Tao proved that H_t (the de Bruijn–Newman deformation of Ξ) has non-real zeros for every $t < 0$. Hence no argument can deduce RH by flowing forward from a real-rooted H_{t_0}, $t_0 < 0$: such an H_{t_0} does not exist. ∎ More generally, ξ sits exactly on the boundary of the Laguerre–Pólya class under this flow, so any proof must be an exact identity rather than an estimate with slack.

7.12 Logical remarks

- RH is equivalent to a Π^0_1 statement, e.g. Lagarias's $\sigma(n) \leqslant H_n + e^{H_n} \log H_n$ for all n. So if RH is unprovable in ZFC it is true, and a proof of its unprovability would require a stronger theory and would establish RH there.
- A detector that is fixed-scale, finitely checkable and sound for off-line zeros at all heights would itself be a proof of RH.
- Perfectly quantized global invariants exist (for example, the norm of the projection onto the orthogonal complement of the Nyman–Beurling subspace is 0 under RH and 1 otherwise, by Bercovici–Foiaş), but they restate RH: finite approximations reintroduce

the detection cost (Burnol's lower bound on the truncated Nyman–Beurling distance decays only like $1/\sqrt{(\log N)}$).

8. Structural measurements

8.1 The archimedean-plus-pole part is not a positive reservoir (Numerical)

Write $W = M + \Sigma E(p)$, where M is the archimedean term plus the pole term plus the constant, and $E(p)$ collects all powers of the prime p. M is indefinite at every window tested, with 2, 2, 3, 4 negative eigenvalues at $X = 3, 5, 8, 12$ (Fourier basis, $N = 40$). Each $E(p)$ is itself indefinite. So at these windows the primes *repair* positivity rather than drain an archimedean reserve, consistent with Suzuki-type numerics in which removing the prime sum makes the lowest eigenvalue strongly negative.

8.2 Positivity is all-or-nothing over the primes (Numerical)

Test: split the primes $\leqslant X$ into teams S, give each team a scalar share α_S of M ($\Sigma \alpha_S = 1$), and ask whether every $Q_S = \alpha_S M + \Sigma_{p\in S} E_p$ can be positive semidefinite. For each team the feasible α form an interval (the lowest eigenvalue is concave in α), so the test is exact up to numerics.

X	primes	best split into smaller teams	worst team margin	all primes in one team
3	2, 3	{2}, {3}	-2×10^{-13} (infeasible, barely)	feasible
8	2, 3, 5, 7	{2,3,5}, {7}	−0.51	feasible*
12	2, 3, 5, 7, 11	{2,3,5,7}, {11}	−0.47	feasible*

*At $X \geqslant 5$ the full team shows values of order -10^{-13}, which is double-precision noise: the true margin is positive but below 10^{-17}.

Positivity appears only when the last prime joins; omitting even the weakest prime costs an order-one amount. The restriction to scalar shares is the natural "no new structure" rule (unrestricted operator splitting makes the question trivial). Consequence: any "local mortar" assembling positivity prime by prime must include genuine cross-prime coupling.

8.3 Function fields: the knife-edge is a blurred exact radical (Proposition + Numerical)

Proposition 8.3. Let C be a curve of genus g over the finite field with q elements, with Frobenius angles ±θ(1), …, ±θ(g), let $\ell = \log q$, and take a window of length $L = K\ell$. Then the window form equals a positive constant times the integral over r from 0 to ℓ of Σ over i of $|\Sigma \text{ over } j = 0, \ldots, K-1 \text{ of } f(r + j\ell)\cdot e^{ij\theta(i)}|^2$, a sum of squares. For each residue r the form on the vector $(f(r + j\ell))$ is the K×K Toeplitz matrix of the Frobenius measure Σ δ at the angles θ(i); it is positive semidefinite of rank at most 2g, with an exactly null space of dimension at least K − 2g.

Proof. All zeros lie at $\gamma = (\theta_i + 2\pi k)/\ell$. Poisson summation in k gives $\Sigma_k |\hat{f}((\theta + 2\pi k)/\ell)|^2 = \ell\, \Sigma_n q(n\ell)\, e^{in\theta}$, with q the autocorrelation of f. Writing $q(n\ell) = \int_0^\ell \Sigma_j f(r + j\ell)\, f(r + (j + n)\ell)\, dr$ and summing over the angles gives the stated sum of squares. ∎

Numerics ($y^2 = x^3 + x + 1$ over F_5: 9 points, a = −3):

window	positive eigenvalues	exactly zero	smallest nonzero
3ℓ	2	1	2.20
6ℓ	2	4	4.71
12ℓ	2	10	11.24

The pole part again has signature (1, 1) and the archimedean-plus-prime part has exactly one positive direction, matching the Hodge structure of §2. But every other direction is *exactly* zero and the margin grows with the window: there is no knife-edge. The mechanism is commensurability: all log N(🪧) are multiples of log q, the zeros repeat periodically, and the window folds into a circle. For ζ, log 2, log 3, log 5, … share no period, nothing folds, and the exact zeros become near-zeros (about 10^{-40} at X = 10). In this language **RH asks why every approximate-radical direction has the right sign**, the job exact vanishing does for curves. The exact-radical statement is presumably known to Connes–Consani in some form; the quantitative side-by-side is recorded here.

8.4 The renormalized margin, and why it cannot be propagated (Numerical + Proposition)

Let $A(X) = \mu_1(X)/\varepsilon(X)$, with $\varepsilon(X) = (2^{14}/3)\cdot\sqrt{2}\cdot\pi^5\cdot e^{-4\pi X + 4.5 \log X}$ the prolate heuristic of Connes (2026, §6.4). For ζ, A is of order one: 2.0 at X = 2, between 3 and 6.6 on $X \in [2.7, 5.4]$, between 8.7 and 11.6 on [10.5, 13.5], trending roughly like $X^{1.2}$ with ±20% wiggles

whose troughs track zeros entering the band (γ_1 near X ≈ 3.1, γ_2 near 3.9, γ_4 near 4.9). Convergence between Fourier basis sizes is 2–7%.

Planted crash (ζ plus one off-line quadruple at T = 30; N = 80, 500-bit):

X	A for ζ	planted, depth 0.01	planted, depth 0.001
4.0	4.4	+54	+54
5.0	6.2	+751	+752
5.5	7.4	+1147	+1335
5.75	7.6	−607	+1909
6.0	6.3	-3.3×10^{4}	+2512
6.5	6.6	-2.5×10^{10}	-7.2×10^{4}
8.0	8.3	-9.5×10^{22}	-5.0×10^{18}

The doomed world's margin *rises* 10–300-fold before collapsing within ΔX ≈ 0.25. Mechanism (Proposition): by the proof of 7.1, $W_{Z_0 \cup Q} \geq W_{Z_0} + R_T - \kappa(\delta, L)\|\cdot\|^2$, where $R_T(f) = |f'(T - i\delta)|^2 + |f'(T + i\delta)|^2$ is positive semidefinite of rank ⩽ 4; the planted points act first as extra zeros, which raise the lowest eigenvalue, and only once the window resolves the depth does the negative part win. **Consequence:** no flow or invariant-region argument on a scalar margin can propagate positivity in the window length; the margin's history is anti-informative about failure. A propagation law must carry the zero configuration in the band, which returns to the canonical-system formulation (Suzuki) and is equivalent to RH.

8.5 Close zero pairs (inconclusive)

Prediction tested: A(X) should dip more when an unusually close pair enters the band. Around the pair $\gamma_{34} = 111.03$, $\gamma_{35} = 111.87$ (gap about 0.37 of the mean spacing), a 0.05-step grid on $X \in [11.70, 12.55]$ shows a largest dip of −8% (at X ≈ 12.05), smaller than single-zero troughs at low X; but two basis sizes disagree by 5–7% there and about six zeros enter per unit of X, so the window margin cannot isolate one pair. A per-zero quantity (the c_k of §6 or $K(\gamma, \gamma)$ of §4) at X ≳ 15 with spectral elements is needed.

8.6 Sensitivity to the prime weights (Numerical; known)

At X = 4, rescaling the weight of the prime 2 by 1% in either direction makes the window form indefinite, so ζ is an isolated point of positivity in prime-weight space at that window. The tolerance scales like the lowest eigenvalue. Sensitivity to the precise value p = 2 is discussed by Connes–Consani (Spectral triples and ζ-cycles).

9. Negative results of the geometric dictionary

9.1 Principal divisors must have degree zero (Theorem + Numerical). In the dictionary "test function = divisor, Connes–Consani radical E(h) = principal divisor", admissible h ($h(0) = \hat{h}(0) = 0$) is not enough. The Mellin transform of E(h) is ζ(s)·Mh(s), which stays finite and nonzero at the poles, so E(h) has nonzero pole pairings. With such E(h), "effective modulo the radical" (D + E(h) ⩾ 0 pointwise) is vacuous: even −H and a strictly negative bump become effective (X = 3, Gaussian-moment generators). Restricting to E(h) with both pole pairings zero makes the notion non-vacuous, and degree-zero E(h) are then trivial for both W and I, as principal divisors should be.

9.2 Naive Riemann–Roch fails (Negative, Numerical). With degree-zero principal divisors and pointwise-nonnegative effectivity, the effective cone does not contain the positive cone {I(D) > 0, degree > 0}. Along $H + s \cdot D_0$ with D_0 ranging over I-eigendirections orthogonal to H, effectivity stops at 45–71% of the positive-cone edge (X = 3). Enlarging the generator family from 20 to 42 changes the ratios by at most 0.02 (0.70→0.71, 0.54→0.54, 0.45→0.47). The correct cone for Weil positivity is positive-definite autocorrelations (Krein), not pointwise-nonnegative functions.

9.3 Fiber classes are not effective, even at the ends (Theorem-level argument + Numerical). $F_1 = I^{-1}a$ is sign-changing, and its best effective representative keeps about 50% negative mass at X = 3–5. Representatives concentrating at the endpoints x → 0, ∞ cannot work: far-separated mass interacts with the window only through the pole kernel, since archimedean and prime kernels decay with distance, whereas F_1 needs I-interaction 1 with F_2.

9.4 Positivity of h is not equivalent to window positivity (Negative, Numerical). Rescaling the p = 2 terms at X = 4 by s ∈ [0.80, 0.98] makes W indefinite (lowest eigenvalue −0.079 to −0.0067) while $h = W^{-1} \cdot \cosh$ stays positive. For s ⩾ 1.01, h changes sign. So "h > 0" is not a criterion. The same experiment shows ζ is an isolated point of positivity in prime-weight space: a 1% change of the p = 2 weight in either direction breaks positivity at X = 4. Sensitivity to the value p = 2 is already discussed by Connes–Consani ("Spectral triples and ζ-cycles").

10. Barrier catalogue and dead routes

This section collects every route the program closed, with the precise reason, so that others need not re-test them. Unless marked otherwise these are recorded kills (arguments or computations), not theorems.

10.1 Five root facts

Every obstruction met in the program traces to one or more of these. (A) Translation adds nothing: shifting test functions does not change the information a window sees. (B) The primes are multiplicatively independent (unique factorization; log p rationally independent). (C) The zeros are random-like and endless, and ξ sits exactly at the boundary of the Laguerre–Pólya class (Λ = 0, Rodgers–Tao): zero slack. (D) Both arithmetic structures are needed (§7.8). (E) Growth is only logarithmic: zero density near height T is $(2\pi)^{-1} \log(T/2\pi)$, so resolving height T needs scales growing with T (§7.1–7.3).

10.2 Fake-world barriers (methods that see only X)

a method that only sees…	fake world it cannot tell apart	status
functional equation + Dirichlet-series shape	Davenport–Heilbronn	classical
a single Euler product + prime number theorem (incl. Selberg's identity)	Diamond–Montgomery–Vorhauer Beurling systems	Proposition 7.7
primes up to X (any window of length log X)	worlds with an off-line quadruple of depth below $\delta_0(L)$, at any height	Theorem 7.1, Corollary 7.3
finitely many values and derivatives of Ξ	Ξ·R with planted non-real zeros	Proposition 7.5
the history of the renormalized margin	a doomed world whose margin is larger until it fails	§8.4
Cauchy–Schwarz with ζ′ weights	an off-line pair with nearly perpendicular values	Proposition 7.6
second moments (pair correlation) with the Montgomery–Taylor window	— (optimal constant for that pure method; Lamzouri via Carneiro–Chandee–Littmann–Milinovich)	known

a method that only sees…	fake world it cannot tell apart	status
prime-sum moments up to order k	independent-phase models matching those moments but breaching positivity at large windows	heuristic (§11.3)

10.3 Nine direct proof attempts and the step each one needs

attempt	the step that fails	what that step really is
1. Li's criterion	the prime part of λ_n cancels down from size about 2^n to size n	no exponential growth, i.e. RH
2. Nyman–Beurling with Möbius mollifiers	$\zeta \cdot M_N \to 1$ in mean on the line	$1/\zeta$ has no poles to the right of the line, i.e. RH
3. CCM spectral triples + Hurwitz	the minimizer ξ_λ stays close to the prolate guess k_λ	needs $\mu_1 \geq -K \cdot \ell(\lambda)$ with $\ell \to 0$, which already forces $\mu_1 \geq 0$, i.e. RH
4. Weil positivity by induction on the window	the Schur complement $D - F^T W(a)^{-1} F \geq 0$	exact positivity on new directions aligned (ratio 1.000) with the near-null space; margins 10^{-5}–10^{-6}
5. de Bruijn–Newman flow with a log-gas energy	each collision costs a definite net amount	a non-smooth quantity computable from primes; smoothing loses the spike
6. Deligne's tensor-power trick	an L_2 with zeros at sums of zeros exists	impossible in the Selberg class (Proposition 7.9)
7. Contradiction through $\psi(x) - x$	an elementary bound of size $C \cdot x^{1/2+\varepsilon}$ on the absolute value of $\psi(x) - x$	square-root cancellation; even a saving $x^{1-\delta}$ would give a zero-free strip
8. Proportion on the line → 100%	mollifier length beyond the off-diagonal limit, and 100% still allows a density-zero exceptional set	an exact sign-change count

attempt	the step that fails	what that step really is
9. Zero-free regions from positive trigonometric polynomials	better polynomials improve only the constant in 1 − c/(log t)^k	positivity deep in the strip, not near the pole

All nine failing steps are one missing object seen from different sides: a structural reason, built from the primes, why a quantity is positive with zero slack.

10.4 Absolute-arithmetic (F_1) routes eliminated

route	kill reason
full addition and full Frobenius on Spec $\mathbb{Z}$ × Spec $\mathbb{Z}$	the square collapses ($\mathbb{Z} \otimes \mathbb{Z} = \mathbb{Z}$; Lorscheid's dichotomy: the self-product is Spec $\mathbb{Z}$ or infinite-dimensional)
THH / derived geometry over the sphere spectrum	richer at each prime, identical rationally; the archimedean room collapses
monoid schemes, blueprints	no cancellation, no cross-factor sums
tropical / idempotent geometry	no signed counts
idempotent Jacobian (Connes–Consani)	heights vanish
Kurokawa tensor products	built from ζ itself (circular)
classical arithmetic locally symmetric spaces	log p is never a geodesic length (Proposition 7.10)
S-arithmetic GL_1	the orbit is dense (Kronecker; root fact B)
Eisenstein series on GL_2 and parabolic blocks	the +1 law (§11.1): ζ enters as ξ(2s), self-adjointness constrains only Re s = 1
natural positive states on the adele class space	wrong sign: states weight primes positively, Weil's form weights them negatively via −ζ′/ζ
period-square formulas	yield nonnegativity of the squared modulus of ζ(½ + it), not Weil's form
Lee–Yang ferromagnetic theorems	the prime couplings are antiferromagnetic

route	kill reason
total positivity / multiplier sequences	ξ is extremal (Λ = 0); the route fails already at low degree

Surviving families at the end of the elimination: an analytic-derived global curve (Fargues–Fontaine at each p plus a twistor-type object at ∞), adelic geometry with signs, Deninger's foliated dynamics, combinatorial/Lorentzian structures, and q-deformations (Habiro-type); plus the non-F_1 spectral programs (CCM, canonical systems).

10.5 A ten-direction parallel search for a gluing principle ("mortar")

Ten independent model instances were each assigned one direction for a principle gluing the existing local two-dimensional structures (one per prime) into a global positive one, and required to pass six tests (Davenport–Heilbronn discrimination, function-field consistency, zero slack, gluing, non-circularity, a numerical prediction). No mechanism survived; every output was an architecture without a discriminating prediction.

direction	outcome
gluing Scholze's local products	sketch; no object or pairing constructed
product formula as positivity source	applies to Davenport–Heilbronn equally; gives cancellation, not positivity
amplification with the frequencies log p	positivity of a square does not transfer to the target form
Rankin–Selberg squaring	classical; reaches Re s = 1 only
statistical mechanics	no model specified; prime couplings antiferromagnetic
Lorentzian / combinatorial Hodge theory	no polynomial constructed
monotone negative index	generic linear algebra; known (Bombieri, Yoshida)
q → 1 limit	no limiting pairing exists
prime-local sums of squares	correct observation that cross-prime terms are required (consistent with §8.2); no certificate
trace formula / operators	the Connes program; same gap

A separate model-generated analysis wrote the prime-power and archimedean source terms exactly as distances from unitary operators (a Blaschke factor per prime, Cayley transforms for the Γ factor). The identities are correct (checked to 10^{-15}) but cosmetic: those terms were already positive, the remaining requirement is Weil positivity, and its proposed completion through Selberg's identity is excluded by Proposition 7.7.

10.6 Every detector pays in scale or precision

For a hypothetical quadruple at height $T \approx 3 \times 10^{12}$ (just above the verified range) and depth $\delta = 0.01$, rough costs of standard detectors:

detector	what must be computed	rough cost
oscillation x^β cos(γ log x) in ψ(x) − x	primes up to x with x^δ ≫ T	$x \gtrsim T^{1/\delta} \approx 10^{1250}$
Mertens function	the same mechanism	the same

detector	what must be computed	rough cost
Li coefficients	until $(1 + \delta/T^2)^n$ dominates (from the mirror zero, whose image under $z = 1 - 1/\rho$ lies outside the unit disk)	$n \gtrsim T^2/\delta \approx 10^{27}$
Speiser (zeros of ζ' left of the line)	derivative zeros near height T	as hard as locating the zero
Weil window positivity	$X \gtrsim T/(4 \log T)$	$X \approx 10^{11}$, precision about 10^{−2×10^{11}}

No detector makes both scale and precision small; §7 makes this rigorous for window tests.

10.7 Earlier arcs

Statements below are reconstructed from July–August 2026 working-session records.

A five-wall obstruction framework. W1, analytic rigidity: by the identity theorem an analytic function is determined by local data, so local perturbation arguments cannot move zeros independently. W2, the explicit formula makes zero locations exactly equivalent to prime error terms, so any bound on primes strong enough for RH is RH. W3, the missing positivity structure, the analogue of the Poincaré-duality/Hodge-index input in Weil's proof for curves. W4, logical status: RH is Π^0_1, so if unprovable it is true. W5, falsifiability asymmetry: a counterexample is a finite check, while a proof has no comparable shortcut. Mapping known approaches against the walls concluded that zero-density and pair-correlation methods are the only live routes avoiding W3, at the price of full exposure to W2.

Proposition 10.7.1 (symmetric-shift impossibility). For real $a \neq 0$, $F_a(s) = \zeta(s + a)\zeta(s - a) = \Pi_p [(1 - p^{-a}p^{-s})(1 - p^{a}p^{-s})]^{-1}$ is an Euler product over exactly the same primes, its completion $\xi(s + a)\xi(s - a)$ is invariant under $s \mapsto 1 - s$, and its zeros $\rho \mp a$ lie off the critical line. Hence no property depending only on the *set* of primes, even together with the $s \mapsto 1 - s$ symmetry, can force zeros onto the line: the weights (the full Euler data, including the critical-line normalization of each local factor) must enter. ∎ This complements Proposition 7.7 and the barrier table of §7.8.

Proposition 10.7.2 (finite-truncation exhaustion). Let $H(a)$ be the admissible test functions of support radius a, split as $H(a) = F \oplus S$ with F a fixed low-mode block and S the increasing union of finite-dimensional high-mode blocks $S(K)$, and let $C(K)$ be the Schur

complement of the Weil form onto F after minimizing over S(K). If C(K) ≥ 0 for every K, then the Weil form is nonnegative on H(a). *Proof.* For f in F and s in S, s lies in some S(K), and f*·C(K)·f, the minimum over S(K) of Q(f + ·), is at most Q(f + s). ∎ The statement is elementary; its use was to reduce support-a positivity to a sequence of finite matrix inequalities. Accompanying facts from the same session: Schur monotonicity C(K+1) ≼ C(K); the minimizers are attained at finite K; the increment series D(∞) = Σ over K of G(K)*·R(K)⁻¹·G(K) converges, with terms of order 1/(K² log K); C(K) was numerically positive for K ≤ 16 at 80 digits for a = log 5. The remaining inequality D(∞) ≼ C(F) for all a > 0 was recorded as Conjecture C. A later 50-digit computation gave a lowest eigenvalue of C(F) near −1.297, contradicting the double-precision runs; since positivity at that support follows from the verified zeros, this points to a formula or normalization error in that computation rather than a counterexample [Likely; unresolved].

Routes killed in that arc: Hardy-space leakage (the archimedean weight ω(t) is negative below t ≈ 6.29); algebraic-Riccati/control formulations (the positive-cost hypothesis is equivalent to the target); quasi-periodic torus methods (A_K⁻¹ accumulates every previous shell, destroying the torus structure); Burnol's local scattering spaces (wrong dimension and index, and nonlinear cross-place mixing from the Schur inverse); zero separation (would need gaps above 1.416, but gaps below 1 occur with positive density); global gap rarity (the operator norm is controlled by the worst local cluster, not the average); Λ as a pointwise Fourier multiplier (the prime comb is a distribution, so a pointwise Φ(ξ) ≥ 0 route does not exist); a Mellin band subspace (entire functions vanishing on an interval vanish identically); a conductor-1 advantage (ζ lacks the log q cushion of higher conductors); and Chebyshev/prime-number-theorem bounds (insensitive to the finite arithmetic perturbations the matrix sees). Also recorded: the Montgomery–Taylor kernel matrix is positive semidefinite on any point set by Bochner's theorem (its Fourier transform is nonnegative); and a heuristic that every reframing tried returned to Weil positivity, which is a summary of experience, not a theorem.

A spectral-universality test (§12.6) showed a product-channel kernel statistic does not distinguish zeta-zero frequencies from controls.

11. Other observations (computations reproduced for this version)

11.1 Eisenstein "+1 law" (exact symbolic computation, block sizes 1–6; reproduced). For the long intertwining operator swapping two Levi blocks of sizes m(A), m(B) with residual (Speh-type) exponents, the constant term is the product over pairs of ξ(υ + a − b)/ξ(υ + a − b + 1), where υ is the difference of the block parameters. Cancelling equal factors in numerator and denominator (Gindikin–Karpelevich telescoping), the surviving

denominator offsets for every pair $1 \leqslant m(A), m(B) \leqslant 6$ are listed exactly in rational arithmetic; for example (1,1): {1}; (1,2): {3/2}; (2,2): {1, 2}; (6,5): {3/2, …, 11/2}; (6,6): {1, …, 6}. The smallest surviving offset is $d_{min} = 1 + |m(A) - m(B)|/2 \geqslant 1$ in every case. Half-integer offsets do occur but always telescope against numerator factors. Hence ζ enters at shifted arguments only, and self-adjointness of the Laplacian (Re $\upsilon = 0$) constrains ζ only on Re $s = 1$ (the prime-number-theorem line), never on the critical line. This is consistent with Lax–Phillips/Faddeev–Pavlov scattering and with the Langlands–Shahidi method reaching exactly Re $s = 1$. The telescoping argument extends to the whole class; a written general proof is not included.

11.2 Nyman–Beurling floors with prime-restricted dilations (Numerical; reproduced). Definition: with ζ sampled on the critical line ($t \in (0, 1000]$, step 0.025, mpmath), weight $w(t) = (\pi(¼ + t^2))^{-1}$, and the Báez-Duarte-type functions $\zeta(½ + it)\cdot n^{-½-it}$, let $D(S)$ be the normalized squared distance $1 - b^T G^{-1} b$ (computed as a weighted least-squares fit of 1) when n ranges over a set S. With S the integers $\leqslant N$ whose prime factors are among the first k primes:

k (primes used)	N = 100	N = 1000
0 (n = 1 only)	0.8577	0.8577
1 ({2})	0.1227	0.1227
2 ({2, 3})	0.0486	0.0486
3	0.0283	0.0280
4	0.0189	0.0181
6	0.0135	0.0118
10	0.0108	0.0082
all primes $\leqslant N$	0.0098	0.0063

The floors are stable in N, so D(k) is a genuine limit for fixed k. One-prime floors: D(2) = 0.1227, D(3) = 0.1766; the two-prime floor D(2, 3) = 0.0486 lies far above the product $D(2)\cdot D(3) = 0.0217$. $D_1 = 0.1227$ matches Szegő's formula for the folded one-prime weight (fold variance 0.1225 plus analytic cost 0.0002). For k = 2 the dual (annihilating) measure carries 51.2% and 35.0% of its energy in the two mixed quadrants ($a < 0 < b$ and $b < 0 < a$ of the 2-torus) and only 1.6% and 0.9% on the negative axes, so the obstruction is genuinely two-dimensional. Allowing rational multipliers 2^a 3^b in [1, 1000] (both signs of

the exponents) lowers the distance to 0.0090, 0.0067 and 0.0059 for |a| ⩽ 18, 30, 48, against 0.0258, 0.0216 and 0.0158 for the same number of random frequencies: the orthant (integer-exponent) restriction is what keeps the floor up. For k ⩾ 2 the winding line t ↦ (2^{−it}, 3^{−it}, …) is dense (log 3/log 2 is irrational), and the remaining floor is an orthant (quarter-plane) prediction problem for a singular measure on the torus, for which no general formula is known. Consistency check: D(all n ⩽ N)·log N = 0.0451 and 0.0434 at N = 100 and 1000, approaching the Burnol–Báez-Duarte constant Σ 1/|ρ|² ≈ 0.0462. Framework: Nikolski's multidisc Hardy-space reformulation and the periodic dilation completeness literature (Dan–Guo).

11.3 Where arithmetic suppression becomes necessary (Numerical + heuristic; reproduced). Let P(t) be the windowed prime sum Σ over prime powers n ⩽ X = e^L of (log p)·n^{−½}(1 − log n/L)·cos(t log n)/π, whose excursions above the budget ρ(t) = log(t/2π)/2π correspond to failures of window positivity. **Verified heights** (window L = 10, primes up to 22026; 60,000–100,000 random heights), real primes versus the same sum with independent uniform phases:

heights	statistic	real primes	independent phases
10^6–$2{\cdot}10^6$	variance	0.389	0.385
	4th moment / var²	2.977	2.966
	6th moment / var³	13.36	14.56
	maximum (budget about 1.91–2.02)	1.83	2.69
	P(P > budget)	0	9.8×10^{-4}
10^5–$2{\cdot}10^5$	6th moment / var³	11.7–12.0	14.8–15.1
	lowest 0.01% quantile	−2.159	−2.159
	highest 0.01% quantile	1.53	2.42
	maximum (budget 1.54–1.65)	1.545	2.81
	P(P > budget)	0	6.2×10^{-3}
	skewness	+0.058	+0.060

The bulk and the entire lower tail agree with independent phases, while the upper tail is cut off exactly at the budget (a one-sided wall), and the sixth moment is 8–21% lower. The deviation appears where theory places the onset of arithmetic correlations: the 2k-th

moment of a short prime sum matches the independent model when $X^k \lesssim T$ (Montgomery–Vaughan); here $X^2/T \approx 500$–5000 (fourth moment borderline) and $X^3/T \approx 10^7$–10^8 (sixth moment deficient).

Beyond verification (heights $[10^{13}, 2\cdot10^{13}]$, budget $\rho = 4.47$): the exact tail of the independent model, from its Bessel-product cumulant generating function by a saddle-point estimate (validated against Monte Carlo at L = 10: 1.13×10^{-3} versus 9.8×10^{-4}, and 6.58×10^{-3} versus 6.0×10^{-3}), gives the expected number of budget crossings on the interval (via Rice's formula):

L	σ	budget in σ	P(S > budget)	expected crossings
8	0.491	9.1	8.6×10^{-24}	2.0×10^{-9}
9	0.556	8.0	1.7×10^{-16}	0.027
10	0.622	7.2	5.8×10^{-13}	81
11	0.687	6.5	9.0×10^{-11}	1.2×10^{4}
12	0.753	5.9	3.1×10^{-9}	3.9×10^{5}
16	1.016	4.4	7.3×10^{-6}	9.0×10^{8}

So at height 10^{13} independent phases would breach window positivity from L ≈ 10 on; for L ⩽ 9 no arithmetic input is needed. At L = 11 and height 10^{13} the real moments still match the independent ones (sixth moment 14.82 versus 14.56; $X^3/T \approx 21$), so the needed suppression lives in rare near-coincidences among products of primes, not in low moments. A localizing-matrix (Hankel) test shows how many moments are needed to certify that the independent model violates the wall: moments up to order 9 at $T = 10^5$ (L = 10, budget 2.6σ), and up to order 29 at $T = 10^{13}$ (L = 11, budget 6.5σ). Tools: Montgomery–Odlyzko, Granville–Lamzouri.

11.4 Davenport–Heilbronn and the negative index (Numerical; reproduced). Setup: $f = ((1 - i\kappa)/2)\cdot L(s, \chi) + ((1 + i\kappa)/2)\cdot L(s, \bar{\chi})$ with χ the character mod 5 with $\chi(2) = i$, and $\kappa = (\sqrt{(10 - 2\sqrt{5})} - 2)/(\sqrt{5} - 1)$, completed by $(5/\pi)^{(s+1)/2}\Gamma((s+1)/2)$. Its off-line zeros at 0.808517 + 85.699348i and 0.650830 + 114.163343i are confirmed ($|f| = 6.5 \times 10^{-7}$ and 2.6×10^{-7} there, against about 0.3 nearby), and its on-line zeros up to height 300 were located by sign changes (204 zeros, 20-digit arithmetic). The window form was evaluated on the zero side, Σ over zeros $\hat{f}(\gamma)\cdot\overline{\hat{f}(\bar{\gamma})}$, in the Fourier basis of a window of length L restricted to frequencies 55–145, with each off-line quadruple entering exactly through its complex points:

window L	basis size	Davenport–Heilbronn: negative eigenvalues (lowest)	control: off-line zeros moved onto the line (lowest)	ζ: negative eigenvalues (lowest)
1	17	0 (+3.69)	+3.69	0 (+1.80)
2	31	0 (+2.76)	+2.76	0 (+0.72)
3	45	0 (+1.37)	+1.55	0 ($+3.2 \times 10^{-6}$)
4	59	1 (−0.709)	+0.221	0 (-1.8×10^{-15})
5	74	2 (−1.84)	$+1.6 \times 10^{-15}$	0 (-2.3×10^{-15})
6	88	2 (−3.63)	-4.9×10^{-15}	0 (-5.4×10^{-15})
8	116	2 (−10.6)	-6.1×10^{-15}	0 (-7.2×10^{-15})

(Values of size 10^{-15} are rounding.) The negative index jumps 0 → 1 → 2 exactly as the window becomes long enough to resolve the two off-line pairs, and stays 0 for ζ and for the control in which the same zeros sit on the line. This is the Bombieri–Yoshida count of off-line pairs seen by a truncation, here observed directly. A complementary local test: on a grid $t \in [82, 90]$, $\sigma \in [0.5, 1]$, $\log|F|$ fails to increase in σ at 69 of 2000 grid steps for Davenport–Heilbronn (all at $t \in [85.44, 85.96]$, beside its off-line zero) and at none for ξ, whose increments at $t = 85.70$ are uniformly positive (0.002 up to 0.038). An earlier test injecting one synthetic off-line pair at height 319 into ζ′s zeros (truncated model, depths 0.05–0.2) gave $\nu = 1$ from the same window for every depth, so in that model the onset was set by the basis size, not the depth; that run was not repeated here.

12. The SR/Rees matrix program and its analytic audits

This section records the program′s other main line: a finite arithmetic matrix ("SR", for signature/Rees) studied as a combinatorial shadow of the Weil form′s signature structure, with Lean 4 formalization, together with roughly 150 numbered research cycles auditing operator, determinant, cohomological and moment-based routes. Sources: the Lean files and the approaches/ ledger of the repository (191 documents), and the author′s progress report (doi:10.5281/zenodo.23174797), which this section corrects where noted. Status labels: **[LEAN]** proved in one of the 16 Lean modules declared in the repository build file

(no sorry, admit or added axiom in any file; build last recorded as verified in commit 444a9be); [LEAN-ARCHIVE] formalized during development, source not in the public repository; **[PROVED]**, **[NUMERICAL]**, **[HEURISTIC]**, **[KILLED]**, **[CONJECTURE]**, **[OPEN]**.

12.1 Definitions

For $X \geq 6$ and $m, n \in \{2, \ldots, X-1\}$, the Rees comparison matrix is $M(m,n) = -1$ if $mn < X$ and $+1$ if $mn \geq X$ **[LEAN]**. Equivalently $M = J - 2B$, where J is the all-ones matrix and $B(m,n) = 1$ when $mn < X$ **[LEAN]**. The quotient count is the number of distinct values $\lfloor (X-1)/m \rfloor$ for m in the support; the threshold matrix S_t on a strictly increasing list of thresholds t has entries -1 when $t_j \leq t_i$ and $+1$ otherwise.

12.2 Lean-verified structure

- **Rank formula** [LEAN]: rank M = (quotient count) $\approx 2\sqrt{X}$; rows m and m' coincide exactly when $\lfloor (X-1)/m \rfloor = \lfloor (X-1)/m' \rfloor$; the threshold representative rows are linearly independent over $\mathbb{Q}$. (The literal $\lfloor X/m \rfloor$ count is not the rank: at $X = 6$ the rank is 2 while that count is 3; and the older claim rank = 2·(square-root cutoff count) fails at $X = 7$.)
- **Threshold matrix** [LEAN]: $S + S^{T} = -2I$; $x^{T}Sx = -\|x\|^2$ for all x; S is injective; $S = -I + K$ with K skew; $\det S = (-1)(-2)^n \neq 0$; rows linearly independent.
- **Global positivity of the all-ones form** [LEAN]: $A(X) = \Sigma\, M(m,n) > 0$ for every $X \geq 6$. The hyperbola reduction $A(X) = (X-2)^2 - 2N(X)$, with $N(X)$ the number of support pairs with $mn < X$, is [LEAN-ARCHIVE]. Benchmarks [NUMERICAL]: $A = 5, 35, 635, 8855, 41381, 5295017$ at $X = 6, 10, 30, 100, 210, 2310$; checked positive for $6 \leq X \leq 1000$.
- **Indefiniteness** [LEAN]: the raw SR quadratic form is not positive semidefinite; global coercivity fails; whenever the form takes a positive value it also takes a negative one; at $X = 6$, e_0 gives a negative value while the constant vector gives 14 (the form is $14c^2$ on constants). The accompanying "no PSD realization" statement is tautological and is recorded as an observation.
- **Dagger structure** [LEAN]: on the (1,1) carrier at $X = 6$, the dagger is a $\mathbb{C}$-antilinear involution swapping labels and fixing the diagonal.
- **Coordinate positivity** [LEAN]: a coordinate functional satisfies $|\Gamma(x)|^2 \geq 0$ and $\mathrm{Re}\ \Gamma(x^{\dagger}x) \geq 0$ on the (1,1) carrier, and is not defined on the (2,0) carrier (a type-level firewall). The Lean names keep the prefix GNS, but these are coordinate norm positivity lemmas, not a GNS reconstruction (see 12.11).
- **Zero transfer** [LEAN]: in a normed ring, $\delta \leq \|F\|$ and $\|G - F\| < \delta$ imply $G \neq 0$, with a pointwise uniform version.

- **Arithmetic bridge** [LEAN]: with S(int) and S(ext) the log-weighted sums Σ log m log n/√(mn) over mn < X and mn ⩾ X, the B3 bound S(int)/T ⩽ (1 − √κ)/2 and κ ∈ [0, 1].
- **Two-channel decomposition** [LEAN]: cos(γ log m)·cos(γ log n) = ½cos(γ log(m/n)) + ½cos(γ log(mn)), splitting the SR observable into ratio and product channels. The ratio-channel diagonal statement (cutoff m² < X, normalized diagonal tending to 1/(2β)) is [LEAN-ARCHIVE].
- **Finite Mellin kernels** [LEAN-ARCHIVE]: sharp cutoff → 1/s, first difference → 1/(s(s + 1)), second difference → 2/(s(s + 1)(s + 2)).
- **Dirichlet series** [LEAN-ARCHIVE]: Σ Λ(n)n^{−s} = −ζ′/ζ and L(1∗1, s) = ζ(s)² for Re s > 1 (one-line applications of Mathlib; not original content).
- **Theta kernel** [LEAN-ARCHIVE]: every corrected theta mode of the Ξ Fourier kernel is strictly positive for t ⩾ 0, and every nonempty finite partial sum is strictly positive.
- **Discrete correlation energy** [LEAN-ARCHIVE]: a finite analogue of the Dimitrov–Xu correlation kernel is nonnegative for nonnegative channel arrays.
- **Placeholders**: the compiled module SR_Mollifier contains four declarations of type True naming open conjectures (SR24_LOG_WEIGHTED_X6_OPEN and three others); they prove nothing and are not counted as results.

12.3 Signature data and the SR Dirichlet series

- **[PROVED]** As X → ∞ the SR Dirichlet series tends to −(ζ(s) − 1)² (support from m = 2), and to −ζ(s)² with full support.
- **[NUMERICAL]** Inertia (p, q, r) of the Rees matrix: (1, 0, 0), (1, 1, 2), (3, 3, 4) at X = 5/2, 6, 30, with monotone negative count; at X = 210, (6, 6, 34) under prime support and (13, 13, 182) under complete support; X = 35 gives (3, 3, 5) under prime support and (5, 4, 24) under complete support. Balance p = q holds at the primorials X = 6, 30, 210, 2310 under complete support; pos − neg ∈ {0, 1} for 6 ⩽ X ⩽ 1000; the radical dominates (87.5% at X = 210); active dimension p + q = 2, 6, 26 at the first primorials. The canonical quotient core has inertia (1,1,0), (2,2,0), (3,3,0), (6,6,0), (9,8,0) at X = 6, 10, 20, 50, 100, invariant under every real basis change (Sylvester).
- **[NUMERICAL]** Densities: balanced even X ≈ 51.6%; balanced X = 2p ≈ 42.3% (0.41–0.43 across ranges), numerically near Conrey's 2/5 with no proven connection.
- **[NUMERICAL]** Spacing ratios of the threshold matrix's nonzero spectrum (cutoff |λ| > 0.01): r = 0.679, 0.284, 0.313, 0.438, 0.542, 0.548 at X = 10, 20, 30, 50, 100, 200, with a large positive outlier; descriptive only, no universality claim.

- **[PROVED]** The exact observable satisfies Q_X = (Σ_{2⩽n<X} w(n))² − 2I_X with w(n) = n^{β−1/2} cos(γ log n) and I_X = Σ_{mn<X} w(m)w(n), explaining its cancellation. At X = 210, Q_total = 1227.86, Q_diag = 3145.27, Q_offdiag = −1917.41 **[NUMERICAL]**.

12.4 Earlier-stage structural findings

A canonical pairing J_X was found to be underdetermined at one coefficient ($b_2(q_{22})$ at X = 5/2, moduli $\mathbb{C}$); dagger compatibility collapses this to $\mathbb{R}$, a five-constraint package forces $J(q_{22}) \in \mathbb{R}_{\geq 0}$, and explicit-formula compatibility pins $J(q_{22}) = (\log 2)^2$. Further recorded statements: raw ordered multiplication is forced; dim E^{20}_6 = 16 (not 10); $q_{23} \neq q_{32}$; the antisymmetric sector has dimension 6; the Rees wall is not an internal dg differential; an all-$\mathbb{N}$ valuation is inconsistent while the positive-domain valuation gives $\mathbb{R}_+$ moduli; the Poincaré-duality pairing has rank one (it factors through L(e_m) = log m) with a three-dimensional radical at X = 6 (witness $\log 3 \cdot e_2 - \log 2 \cdot e_3$), positive definite only on the quotient by ker L; and a proposed partial sign theorem on ker L is false (an explicit counterexample with positive Rees-signed square). The Rees sign convention (interior ↦ −1) is Hodge-index motivated but not derived from the explicit formula, which was recorded as the open problem of that stage.

12.5 The Hardy–Gram kernel

Let G_H(m, n) = (√(mn)/2π) ∫ (m/n)^{it} |ζ(½ + it)|²/(¼ + t²) dt.

- **[PROVED]** Exact multiplicativity G_H(cm, cn) = c·G_H(m, n), hence G_H(m, n) = gcd(m, n)·G_H(m/g, n/g); and G_H(m, n) = √(mn)·H(log(m/n)) with H the Fourier transform of |ζ(½ + it)|²/(¼ + t²).
- **[HEURISTIC]** From the approximate functional equation, G_H(m, n)/G_H(1, 1) → gcd(m, n) as the cutoff grows. **[NUMERICAL]** The ratio G_H(1, 2)/G_H(1, 1) decreases only from 1.230 (T = 100) to 1.204 (T = 1000), so convergence is not yet supported (Conjecture 12.10.4).
- **[NUMERICAL]** With the gcd kernel, the Weil-form contraction fails, with ‖C(L)‖ − 1 ∼ L^{7.30} (4-point power-law fit, L ∈ [6.5, 8.0]); the dominant negative eigenvector of G(L)*G(L) − F(L)*F(L) concentrates on prime-power channels n = 2, 4, 7, 9, 13, 16, 17, 19 for L = 6, 7, 8, not on boundary primes near exp(2L).

12.6 Exact failure theorems for specific routes

- **SR coefficients are not von Mangoldt weights** [PROVED]: the SR coefficient at (6, 4) is −1 while Λ(4) = log 2; SR produces divisor-count coefficients.

- **Abel bridge at the wrong scale** [PROVED]: S_int(X) ~ 2√X log X while $2\psi_2(X)/\sqrt{X} \sim X^{3/2}$.
- **ζ^2 gives zeros, not poles** [PROVED]: the full-support limit $-\zeta(s)^2$ has double zeros at the zeta zeros, so a Perron/explicit-formula argument receives no contribution from them.
- **A clamped Weil/Paley–Wiener inequality fails at small L** [PROVED]: for $h(y) = y(1 - y^2)^2$, $R_L \to 337/105$ while $c_L \sim \pi^2/(2L)$; for every $0 < L \leqslant 1/10$, $R_L < 4$ and $c_L > 20$.
- **Relaxed-source scaling** [PROVED]: for $D = c \cdot h$ and any $\lambda > 0$, $\lambda^2(e - \ell^2) + 8\lambda\ell - 16 \leqslant 0$ fails for large λ whenever $e > \ell^2$; contractivity cannot extend to arbitrarily scaled sources.
- **Volterra endpoint mismatch** [PROVED]: the causal convolution representation vanishes on terminal boundary mass ($V_{\delta_T} = 0$ on $L^2(0, T)$) although the cosine-channel difference is nonzero; smallest counterexample $\alpha = \delta_T$ with $0 < M \leqslant (\log 2)/2$.
- **Finite moment closure is impossible** [PROVED]: a polynomial identity closing the terminal moment ladder at finite order would vanish on a continuum, so it is the zero polynomial; the exact scale/endpoint state is infinite-dimensional.
- **Ratio-channel GUE detection is γ-independent** [PROVED]: $M(m, n) \cdot e^{i\gamma(\log m - \log n)}$ is unitarily equivalent to M by diagonal conjugation, so its spectrum is independent of γ. The product channel is non-specific [NUMERICAL]: at X = 200 the non-zero frequency $\gamma = 35$ gives r = 0.6054, closer to the GUE value 0.5996 than any tested zeta zero.
- **The Möbius–log diagonal realization is indefinite** [PROVED]: $\mu(2) \log 2 < 0$ and $\mu(30) \log 30 < 0$. (Narrow statement: trace alone does not determine the signature of other realizations.)
- **The Báez-Duarte target in the gcd model is tautological** [PROVED]: with $b_k = 1/\sqrt{k}$, b is the first column of G, so $b^T G^{-1} b = 1$ and $d_N = 0$ for every N by linear algebra.
- **The Arakelov-type matrix is strongly indefinite** [NUMERICAL]: $\tfrac{1}{2} \log(\mathrm{lcm}/\gcd^2) \cdot \gcd/\sqrt{(mn)}$ has minimum eigenvalue ≈ -5.49 with 99 negative eigenvalues at N = 100.
- **Standalone SR mollifier** [KILLED]: κ_{SR} decreases from about 0.763 at X = 6 to 0.224 at X = 2310; no uniform 75% consequence.
- **Conrey exponent as an SR growth law** [KILLED]: the predicted slope $3/2 + (2/5)(\beta - \tfrac{1}{2})$ is refuted (observed 2.850 versus predicted 1.620 at $\beta = 0.8$, $\gamma = 21.022$) once the exact product-of-cosines definition is used.
- **Further dead ends** [KILLED]: the original B3 with $\psi_2(X)/X^2$ (false at X = 6, c = ¾); a mixed log/von Mangoldt Cauchy–Schwarz (different weight vectors); a direct scalar-mollifier reading of the two-variable Rees kernel; the Weyl shortcut $\mathrm{pos}(J - 2A) \leqslant 1$

(A_X is not positive semidefinite); "diagonal equals reesTrace" (cutoffs $m^2 < X$ versus $2m < X$: 20 versus 2 at $X = 30$); a pointwise row bound $|\mathrm{RowSum}_m| \leqslant (X/m)^{\beta}/\beta$ (row 4 versus 2.19 at $X = 6$, $m = 5$); a two-sided growth law $C \cdot X^{3\beta-\varepsilon} \leqslant |Q_X|$ (phase oscillation forces arbitrarily small amplitudes); and the Möbius-sieve claim $\sum \mu(n)(d(n) - 2)n^{-s} = 2/\zeta(s)$ (false; the correct series is $\zeta(s)(\zeta(s) - 2)$ for $a(1) = -1$, $a(n) = d(n) - 2$).

12.7 Structural obstructions

- **Type mismatch** [PROVED, narrow]: RH is a support condition (zeros on a line), matrix positivity a sign condition; this particular SR matrix has no proved mechanism converting finite sign information into the analytic support statement (other finite matrices, e.g. Redheffer's or Li's, do encode RH-equivalent data).
- **Separability** [PROVED]: $\Lambda(mn) \neq \Lambda(m)\Lambda(n)$, so no separable kernel $f(m)g(n)$ produces Λ as a coefficient sequence.
- **Circularity** [PROVED, narrow]: the construction's Nyman–Beurling target requires critical-line values of ζ in a way that depends on the zero locations.
- **Arithmetic/spectral duality** [PROVED]: the explicit formula's prime side and zero side are dual descriptions of one functional, not independent positive energies; adding zero-side energy changes the form instead of improving it.
- **Self-adjoint Euler coupling** (Cycle 73) [PROVED]: for real diagonal D with entries < 1 and nonzero real symmetric K with zero diagonal, $\det(I - D - \varepsilon K)$ changes at second order in ε, so a symmetric coupling cannot preserve the Euler determinant.
- **Coercivity versus determinant zeros** (Cycle 36) [PROVED]: since $S + S^{T} = -2I$, every eigenvalue λ of S has $\mathrm{Re}\,\lambda = -1$, so the coercive SR block cannot itself carry determinant zeros.
- **Self-adjoint Schur cancellation transfers no positivity** (Cycle 42) [PROVED]: $K = [[S, B^*], [B, BS^{-1}B^* + D]]$ is congruent to $S \oplus D$, so its inertia is that of S plus that of D.
- **Prime-loop gauge obstruction** (Cycle 53) [PROVED]: a two-state loop $[[0, a], [b, 0]]$ preserves the Euler factor $1 - p^{-s}$ iff $ab = p^{-s}$; any sign gauge leaves the determinant unchanged, so signs cannot create positivity. Positivity versus primitive cycles (Cycle 56): for nonnegative weights the mixed-cycle correction $-uv$ in $\det(I - [[a, u], [v, b]]) = (1 - a)(1 - b) - uv$ is strictly nonzero when both couplings are present; engineered (Cycle 57) and graded/supersymmetric (Cycles 58–59, 82–83) cancellations either change the Euler determinant or fail to convert into an ordinary positive determinant.

- **Mayer sign-cocycle gauge theorem** (Cycle 77) [PROVED]: on a strongly connected transfer graph, if a sign modification σ has the same product on every directed cycle, then $\sigma_{ij} = \upsilon_i \upsilon_j^{-1}$ for vertex signs υ, so the modified transfer is conjugate to the original by a diagonal sign matrix and every determinant and spectrum is unchanged. A sign insertion with a non-trivial cycle product changes the trace-log and hence the determinant. So every determinant-preserving sign insertion is spectrally invisible; non-trivial skew products decompose over the group's representations (Cycles 78–79).
- **Multiplication-shift acyclicity** (Cycles 50, 80): forward multiplication-by-prime operators are nilpotent on finite truncations (no cycles, trivial determinant); adding reverse edges creates composite cycles that destroy the Euler product (Cycle 81).
- **Positive heat trace versus Möbius** (Cycle 86): $H(t) = \Sigma\, \mu(n) e^{-nt}$ has coefficients 1, −1, −1, 0, −1, 1, …, so it is not the heat trace of any positive self-adjoint operator; the divisor-complex construction (reduced Euler characteristic of the proper-divisor order complex equals μ(n), Cycles 61–69, with Mellin transform $1/\zeta(s)$ for $\mathrm{Re}\, s > 1$) yields the Möbius signs only as a supertrace.
- **Scalar and low-rank repairs** (Cycles 89–90): the scalar shift α_X needed to make $M_X + \alpha_X I$ positive semidefinite grows from 5.94 ($X = 20$) to 40.03 ($X = 300$), with $\alpha_X/\sqrt{X}$ increasing from 1.33 to 2.31, so no bounded archimedean diagonal term works [NUMERICAL]. A positive update $M + UU^* \succeq 0$ needs rank $U \geqslant \nu_-(M)$ (the negative index) [PROVED], and $\nu_-(M_X)$ grows from 3 ($X = 20$) to 18 ($X = 400$) [NUMERICAL], so no fixed-rank archimedean sector can repair the form.
- **Metric identifiability** (Cycle 97) [PROVED]: a finite symmetric form W does not determine the operator metric; every positive definite G gives a different generalized operator $G^{-1/2} W G^{-1/2}$. An ordinary common L^2 embedding of scale-dependent bases fails (Cycle 96), and Sobolev weights $(1 + n^2)^{a/2}$ do not cure the form's degeneration (Cycle 98).
- **Finite Euler determinants have the wrong zero geometry** (Cycle 127): $1 - p^{-s}$ vanishes on $\mathrm{Re}\, s = 0$; shifting to $1 - p^{1/2-s}$ puts zeros on the critical line but at the wrong ordinates (Cycle 128).
- **Positive definiteness does not force real zeros** (Cycle 146): $2 + \cos z$ is the Fourier transform of a positive measure yet has non-real zeros; the correct target is total positivity / Laguerre–Pólya criteria (Dimitrov–Xu correlation-density conditions, Cycles 147–148).

12.8 Further research cycles

- **Mayer transfer operators** (Cycles 22–29): Mayer's identity $D(q) = \det(I - L_{q/2}) = \zeta(q-1)/\zeta(q)$, the telescoping $\Pi_{j\leq N} D(q-j) = \zeta(q-N-1)/\zeta(q)$, functional-equation normalization, and a trace-class realization of the archimedean factor through the Weierstrass product of $1/\Gamma$; a near miss, with no positivity.
- **Completed-ξ targets** (Cycles 30–34): determinant positivity is impossible for $1/\xi$; a two-dimensional indefinite pencil $[[0, 1], [-f, 0]]$ linearizes $f = 1/\xi$; Li coefficients were computed as a Weil-type replacement.
- **Hilbert–Pólya separation and Taylor companions** (Cycles 84–85): diagonal matrices of known ordinates and companion matrices of truncated $\xi(\frac{1}{2} + z)$ reproduce zeros only by construction.
- **Finite arithmetic Weil operator** (Cycles 93–116): a finite Weil matrix built only from $\Lambda(k)$, $\log k$ and Fourier test functions; a rank-one Weil-scaling operator self-adjoint in the ground-state-shifted Weil quotient, whose regularized determinant is the Fourier transform of the selected ground state (the Connes–Consani–Moscovici construction, reproduced independently); larger-cutoff scans at $\lambda = 2, 3, 4$ reproducing 14.1347…; form-convergence and Mosco-convergence routes for the ground states, with the single sufficient global theorem isolated (convergence of the selected Weil ground states), and a fixed-scale infinite operator on $L^2_{\text{even}}([-\log \lambda, \log \lambda])$.
- **Euler log-determinants and completed ξ** (Cycles 123–126): the exact logarithmic-derivative bridge to Λ, a positive diagonal Euler operator for real $s > 1$, and convergence of finite completed Euler products toward ξ in the half-plane of convergence.
- **Jensen and Hankel tests** (Cycles 129–136): odd Taylor coefficients of $\xi(\frac{1}{2} + z)$ vanish to 80 digits; Jensen polynomials of Ξ are real-rooted through degree 12 at the origin and for shifts 1–4 at degree 6; raw Taylor Hankel matrices fail to be positive, but with the Fourier-moment normalization $c_n = (2n)!\, b_n$ the Hankel matrices are positive through size 7 (minimum eigenvalues 1.90×10^{-3}, 2.91×10^{-5}, 7.47×10^{-7}, 2.41×10^{-8}, … for sizes 2–5).
- **Theta kernel** (Cycles 137–143): an earlier spurious kernel negativity (from evaluating a one-sided series at negative arguments) was corrected; individual theta modes and the total kernel are positive on the half-line (Lean-verified, 12.2), with Gaussian tail convergence.

12.9 What a successful operator would need

Six properties were formalized for an operator T(s) that could prove RH: P1 det(I − T(s)) vanishes at the zeta zeros; P2 traces of powers count prime powers with weight Λ; P3 a canonical quadratic form is positive; P4 T(s) and T(1 − s) are related with Re s = ½ as fixed line; P5 arithmetic origin with finite truncations; P6 the determinant is well defined and continues analytically through the strip. No examined operator has all six: the SR Rees matrix and its full-support and threshold variants have only P5 (the threshold matrix also P3); $-D^2 + \frac{1}{4}$ on the Weil/Paley–Wiener side has P4 and P6 with partial P1 and P3; the Mayer transfer operator has P1, P2, P4, P6 but not P3 or P5; Connes′ adelic space has P1, P2, P4, P6 with P3 open; the gcd matrix has P2, P3, P5 with partial P1; the normalized Hardy–Gram kernel has P1 and P3 with the rest partial.

12.10 Conjectures and the frontier

1. **Primorial balance** [CONJECTURE]: p = q for the Rees matrix at every primorial X under complete integer support (verified X = 6, 30, 210, 2310).
2. **Universal signature bound** [CONJECTURE]: |pos(X) − neg(X)| ⩽ 1 for all X ⩾ 6 (verified X ⩽ 1000); interlacing controls each inertia count separately but lets the gap accumulate, so it does not prove the bound.
3. **Dagger-completion defect** [CONJECTURE]: the common obstruction across the SR, Connes and Weil programs is the inability to convert forward multiplicative structure (A^2) into Hermitian structure ($A^\dagger A$); equivalently, the (2,0) + (0,2) versus (1,1) Hodge obstruction. Not formalized; novelty relative to Connes (1999) is unchecked.
4. **Hardy–Gram gcd asymptotics** [CONJECTURE]: $G_H(m, n)/G_H(1, 1) \to \gcd(m, n)$, plausibly from the twisted second moment of ζ.
5. **Candidate equivalences** [CONJECTURE, none proved]: positivity of a normalized von Mangoldt SR Dirichlet form for every cutoff and test polynomial; convergence of the primorial Rees strand to 1 from above with uniform control of the other strands; positivity of the explicit-formula transform of the Λ-weighted Rees kernel; and $S(P_k)/(P_k^2/2) \to 1$, linking the log-weighted interior sum to Chebyshev's ψ_2.

Frontier [OPEN]: construct, from the completed arithmetic source, a contractive system whose first derivative maps a structurally derived input to the observed boundary state with input norm exactly |4 − L| (a generic Schur factorization with these properties is merely equivalent and not evidence); certify or kill the Hardy–Gram asymptotics; compute the Nyman–Beurling target analytically without RH; the last step, convergence of the Nyman–Beurling approximation, is RH itself (Báez-Duarte). The working hypothesis recorded across the program: the critical line is enforced by a global self-duality plus

positivity mechanism, not by a local Euler factor or a finite sign matrix, and every proposed construction must answer which exact symmetry prevents a negative direction of the global Weil form from becoming an off-line zero.

12.11 Corrections to the earlier paper

The author's earlier report contains errors corrected here: it stated $\mu(2)\log 2 > 0$ (in fact $\mu(2) = -1$); its "no positive-semidefinite realization" theorem is tautological (no everywhere-nonnegative function equals one taking negative values) and is relabeled an observation; several files contained statements of the form $A \to \text{True}$, which compile but prove nothing, and were removed; "GNS positivity" reduces to $|x_{22}|^2 \geq 0$ and is a coordinate norm lemma; the L-series identity for Λ is a one-line application of Mathlib, not an original contribution; and its acknowledgments section has been rewritten (the research directions, including the ledger structure, the dagger-completion hypothesis and the FICS methodology, are the author's). Documented repository anti-patterns: proxy drift (threshold matrix versus Rees matrix), semantic inflation of names, compilation treated as the objective, library-result laundering, and finite-to-analytic slippage without a proved bridge.

12.12 The operator-candidate search

The program's operator ledger records 297 named candidate operators (A–Z, AA–JH, with a second CH–CZ series and a DA2–EF2 series), 124 cycle files, 11 failure records and 34 general research files. Each candidate was scored against the operator criteria of 12.9 (in later entries: a multiplicative/additive bridge, a fixed positive operator, functional-equation symmetry, a determinant identity with Ξ, and a cutoff limit). **No candidate satisfies all of them.** The families, with representative verdicts:

- **Finite Euler, Blaschke and companion realizations** (e.g. B, V, W, X, HW, HX, EX): finite Euler/Blaschke zero geometry instead of Ξ; the companion matrix of a finite Euler numerator has the right determinant but is neither self-adjoint nor contractive (norms 2.867, 4.040, 6.367 for three, four and six prime factors); a positive diagonal Euler contraction cannot carry reciprocal symmetry, which would require eigenvalues $\exp(p/10) > 1$.
- **Indefinite (Krein) and Cayley repairs** (e.g. Y, Z, DS, DT, HQ, HS): the reciprocal pair $\mathrm{diag}(a, a^{-1})$ is J-self-adjoint with determinant $(z - a)(z - a^{-1})$, but its Hilbert norm is $a^{-1} > 1$ and JA has a negative direction; the Cayley transform makes it unitary, but its determinant is only a finite Cayley factor ($\det U = -1$ for each reciprocal pair).

- **Rank-one and low-rank couplings** (e.g. AA, AB, BN, S, EB2): non-factorized determinants need a cutoff-dependent normalization; with the arithmetic normalization $\alpha_P = 1/(1 + \Sigma \log p \text{ over } p \leqslant P)$ the norm exceeds 1 at small cutoffs, the coupling vanishes as P grows, and the smallest eigenvalue collapses.
- **Prime–archimedean couplings** (e.g. AD–AH, AY, BF, DW2–DZ2, IF): direct sums factorize; coupled versions are positive and contractive through P = 160, but nothing arithmetic fixes their scale (the heat coupling's raw norm grows from 2.540 at cutoff 10 to 5.567 at cutoff 80, forcing the admissible scale below 0.180). DY2 reduces finite positivity exactly to the Schur contraction $\|P^{-1/2} W G^{-1/2}\| \leqslant 1$; uniformity in the cutoff is the open step.
- **Exact structural bridges without analytic identification** (AI–AR): a divisor-convolution Hopf structure, a finite Heisenberg–Weyl algebra (relations hold below 10^{-14}), logarithmic prime translations on $L^2(\mathbb{R})$, the Bohr compactification, a profinite odometer, Cuntz dilations ($L^*L = (\Sigma |a_p|^2)\cdot I$, branch norm about 0.387) and a positive Ruelle transfer. Each supplies an exact multiplicative–additive or positivity structure; none produces Ξ.
- **Theta-kernel operators** (e.g. AS–AW, BY–CT, CU–DA, CG): convolution by the even theta kernel Φ is multiplication by Ξ(t) under Fourier transform, and a rank-one pencil gives $\det(I - T(z)) = 2\xi(\tfrac{1}{2} + z)$ exactly; but the positivity is that of Φ, which does not force real zeros (12.7: 2 + cos z), and these identities are built from ξ itself.
- **Weil-form compressions and signature normalizations** (e.g. AX, AZ, BB–BE, CV, CW, DB, DH, DK, DO, GC): for any nonsingular finite Hermitian compression, a congruence turns Q_X into $\operatorname{diag}(\operatorname{sign} \lambda_j)$ exactly, so signature normalization carries no information by itself; certified CCM-type compressions reproduce known positivity.
- **Pick, Herglotz, Stieltjes and Hankel moment realizations** (e.g. BP–BX, CE, FD–FI): positivity of the moment and Pick kernels holds for the arithmetic measures used, but determines no completed determinant.
- **de Branges spaces, canonical systems and Krein strings** (e.g. DC, DL, DM, EM, EN, IB–IE, IY): the Hermite–Biehler companion is not derived independently from the prime and gamma data; constructing it is the canonical-system formulation of RH.
- **Scattering, colligations and transfer systems** (e.g. DN, EO–ET, IG–IP): local passive or conservative realizations exist; the global inner/outer factorization is the open step.
- **Quantum graphs and Ihara-type determinants** (DN2–DV2, EU): cycle lengths log p are exact, but the von Mangoldt amplitudes and the additive part are not canonical.

- **Determinant lines, orientations, Pfaffians and K-theory** (e.g. CC, CD, GN–GV, HK, HL, HT, HU, HY, HZ, IL, IV): positivity of a squared determinant loses exactly the phase/orientation information that separates zeros on and off the line.
- **Semigroups, Dirichlet forms and heat traces** (e.g. BM, BU, FU–FW, GD–GG, GK–GM, GW, HV, IS, IW): positive Lévy-type and Dirichlet-form structures exist for Λ, but heat traces of positive operators cannot carry Möbius signs (Cycle 86).
- **Adelic, cocycle and connection constructions** (IQ–IU, IZ–JH): local assembly is formal; the global trace identity and the product-formula holonomy normalization are unproved.

The recurring pattern is a trade between criteria: positivity and contraction against reciprocal (functional-equation) symmetry; non-factorization against cutoff stability; exact structure against analytic identification with Ξ. Each trade is an instance of a barrier in §7–§10. The complete ledger (about 470 kB) is published as supplementary/OPERATOR_SUMMARY.md in the companion repository github.com/loganeberwein/Logan-Eberwein-RH-Weil-Window.

12.13 Archive searches

Ten further search ledgers (about 5.5 MB of records) document structured searches run under fixed protocols, each candidate recorded with its construction, its tests and an exact failure point. They are published as supplementary/archive_searches/ in the companion repository. Summary (counts are of indexed candidates or passes as recorded in each ledger):

ledger	target	size	outcome
Frobenius Search, Gate 1	a geometric Frobenius flow whose primitive closed orbits have lengths log p	54 candidates, 8 failure classes	proved: universal circle-gap collapse for unitary flat twists; partial: trace divergence from one orbit per prime under positive smearing, spectral filtering changes orbit coefficients, exact cohomological cancellation erases orbit contributions, spectral-scale versus orbit-length incompatibility
Frobenius Search, Gate 1 (v3)	global prime-orbit operators	133 candidates (33 explicitly killed)	spectral-only global encodings keep a length list but lose closed-orbit and Lefschetz data
Gate 1A	a zero-independent Hilbert-space realization	27 candidates, 25 failure classes	proved: continuous-base compact-fiber obstruction, polynomial-confinement noncompactness, compact-circle faithfulness loss, cylinder-base persistence; partial: strip-multiplication noncompactness, Paley–Wiener strip-width limit, discrete-sampling invisibility, among others

ledger	target	size	outcome
Genius Search protocol v2	discriminating ideas tested against toy models (finite fields, lattices, couplings)	121 indexed ideas (29 recorded dead, 9 partial in the summary tables)	no idea passed all discrimination tests; a seed bank and operator scorecard are recorded
Quadratic Positivity Search	exact splittings $Q(h) = P(h) + R(h)$ with $P \geqslant 0$ for a structural reason and R controlled	16 passes, 14 failure classes	each splitting fails through an uncontrolled remainder (Poisson, Euler-translation, twisted comb, theta, log-derivative cross-terms, character-orthogonality, Haar diagonal loss, Möbius–divisor lifts, truncated Euler inverse, uncoupled Rankin–Selberg norm); a partial no-go covers positive-coefficient cross-term matching
Realization Functor Search	a functor from arithmetic correspondences to a cohomology carrying a Lefschetz pairing (e.g. Beilinson–Parshin adelic divisors)	82 attempts	each records the exact obstruction to mapping degree-zero correspondences to primitive classes with intersection pairing equal to the Lefschetz pairing

ledger	target	size	outcome
Region Search	exact constraints on the factorization of n and n + h in which addition and multiplication interact	83 indexed ideas	exact identities only, e.g. $D(n(n+h)) - D(n^2) - D(nh) = n(D(n+h) - D(n) - D(h))$ for the arithmetic derivative D, $\gcd(n(n+h), 2n+h)$ divides h^2, and a radical-ratio criterion forcing $\mu(n)\mu(n+1) = 0$; a scan to 10^7 (radical-ratio mean 0.5286, maximum 1.5679 at n = 4374); no new milestone
Positive Prime Characterization Search	characterizations of primality by positive structures (divisor posets, incidence Gram matrices, and others)	166 candidates	recorded criterion by criterion
Positive Scale-Separation Search	positive statistics isolating prime information across scales	42 passes, 27 failure classes	proved: kernel-shape restrictions for positive Chebyshev smoothings and short-window differences, and pointwise blindness of truncated divisor-feature sieves (rough composites look prime); positive higher-order divisor weights, lcm-Gram energies, norm-lattice transforms and factorial valuations all mix scales or leave the aggregate unevaluated

ledger	target	size	outcome
Ground-State Simplicity and Parity Search	simplicity and parity of the localized Weil ground state	22 failure classes (numerical CCM truncations through 28 passes)	proved: a signed-triangle obstruction to multiplication gauges, near-diagonal failure of total positivity, a dilation-covariance obstruction, a quadratic Sturm–Liouville commutant mismatch, and a high-shift resolvent sign obstruction at a = 0.17; finite-section diagnostics leave the cutoff-free parity ordering open

These ledgers are recorded at the level of the summaries above; individual proofs are in the files and have not been independently re-verified for this paper.

13. Retracted claims and pitfalls

Recorded so that others do not repeat them.

claim we made	what was wrong	correction
Arithmetic factor $A(\lambda) \approx 74$–173 in a Weil–Slepian factorization	mis-normalized Slepian asymptotic; Fourier truncation ~20% off	$A(\lambda) \approx 4$–6 with exact leakage and spectral elements
Weil's form is the (negated) intersection form	it is positive semidefinite, but an intersection form has one positive direction	W is the Castelnuovo–Severi expression; the intersection form is the archimedean-plus-prime part (§2)
Fibers become effective by moving mass to the endpoints	endpoint mass has no archimedean/prime interaction	fibers are not effective (§9.3)

claim we made	what was wrong	correction
Effectivity modulo the radical works for fibers	notion was vacuous without degree-zero restriction	§9.1
Detection law $\delta \cdot L^2 \approx 6$ is general	holds only above the band edge, roughly 6–10	§5
A certified window gives an off-line-zero exclusion region	the conversion cancels at order δ^2	δ^* is counterfactual only (§5)
Half-space Nyman–Beurling relaxation solved via Helson–Lowdenslager	the recalled theorem did not apply as stated	treat both half-space and orthant as needing proof
Mollifier length is the barrier to 100% on the line	measured: not the binding obstruction	—
$\nu(L)$ onset follows the detection law	onset set by window degrees of freedom in the truncated model	§11.4
The archimedean-plus-pole part is a positive reservoir drained by the primes	it is indefinite at every window tested	the primes repair positivity (§8.1)
Positivity of the ground state h is a criterion for window positivity	h stays positive in perturbed models whose form is indefinite	§9.4
Difficulty can be dispersed evenly over window lengths and propagated by a flow on the renormalized margin	per-step conditions are independent, and a doomed world's margin rises before failure	§8.4
Close zero pairs produce visibly deeper margin dips	not resolvable at $X \approx 12$: basis error equals the effect, several zeros enter per unit X	inconclusive (§8.5)
A quantized global invariant (Nyman–Beurling projection norm) is a fixed-scale detector	it is a restatement of RH; finite approximations reintroduce the detection cost	§7.12

claim we made	what was wrong	correction
A completion of squares through Selberg's identity can glue the prime channels	the identity holds in every Beurling system, including ones violating the RH analogue	§7.7
Removing RH from Conrey–Ghosh–Gonek mirrors the unconditional Montgomery argument	one off-line pair inflates the Cauchy–Schwarz ratio without bound	§7.6
A prepared Lean bound of 0.6727 for the critical-line proportion would set a record	the public leaderboard record had moved to 67.34812% (eight-point weighted rank-trace refinements)	not submitted

Numerical pitfalls.

- The constant c_0 must be evaluated with $(X+1)/(X-1)$ in working precision; a double-precision ratio shifts every eigenvalue by a constant (observed: -6.3×10^{-17}).
- On diagonal panel blocks the $-q(0)$ subtraction in the archimedean integrand persists across the entire y-range; dropping its no-overlap tail produced spurious negative Ritz values of order -0.25.
- y-panels must be subdivided by oscillation count, not only at the breakpoints $\log n$ and $L - \log n$; a long panel near $[0.069, 0.230]$ at $X = 3$ was under-resolved at quadrature order 10.
- Global Fourier bases are inadequate beyond $X \approx 6$ for eigenvalues: near-null eigenvectors have weak singularities at every $\log n$ and $L - \log n$, and graded spectral elements converge exponentially.
- A finite archimedean cutoff T produces spurious negative eigenvalues in deep spectra; trust only digits stable under a change of T (see arXiv 2607.02828).

14. Open problems and references

Open problems.

1. Prove the operator version of the inertia identity (§2), with Fredholm conditions on I_L.
2. Bound the error between $W^{-1}\cdot\cosh$ and the ground state uniformly in L, and decide whether $\hat{h}$ has only real zeros.

3. Prove $K_L(\gamma, \gamma) \to 1$ exponentially for in-band zeros, which needs a quantitative near-null-space theory.
4. Derive the band edge $T_b(X)$ (empirically $\approx 4X \log X$) and the in-band slope from prolate/Landau–Widom asymptotics.
5. Explain the per-zero constants c_k, for example through neighbouring gaps.
6. Find the right cone for a Riemann–Roch-type effectivity statement (positive-definite rather than pointwise), and test whether it contains the positive cone of I.
7. Certify the exact-form values of Theorem 7.2 (now computed, §7.2) with interval arithmetic and a basis-truncation bound, and give a clean asymptotic proof of where window blindness sets in (the empirical band edge $4X \log X$), presumably via sampling estimates for the zeros near height T.
8. Check that the hypotheses of Gabardo's distributional extension theorem cover the explicit-formula distribution exactly as used in Corollary 7.3 (prime deltas and the archimedean principal-value term).
9. Make the Chirre–Gonçalves–de Laat semidefinite-programming bound (0.6792 under RH) unconditional through the off-line-pair inertia treatment; decide which of its positivity constraints survive.
10. Find a dictionary between window length X and the degree/shift of Jensen polynomials, so that the Griffin–Ono–Rolen–Zagier asymptotic theorem speaks about windows, or conversely.
11. Settle the Primorial Balance Conjecture and the universal signature bound of §12.

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